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Wind-Induced Vibration Control

12.1 Preview

It has been discussed in previous chapters that long-span cable-supported bridges are susceptible to flutter instability, vortex-induced vibration, buffeting-induced vibration, vehicle-wind-bridge interaction, multiple loading-induced fatigue and rain-wind-induced cable vibration. To ensure the safety and functionality of bridges and vehicles under wind loading, it is necessary in some cases to find efficient and effective measures for mitigating excessive wind-induced vibration to a acceptable level, and for improving bridge stability to avoid collapse. In general, there are three ways to mitigate wind-induced vibration of a long-span bridge: structural modification, aerodynamic measures and mechanical measures. The realization of these mitigation technologies may also make the construction of super-long-span bridges come true and elevate bridge design concepts from a static and passive level to one with dynamism and adaptability.

A brief introduction on wind-induced vibration control methods will be first given in this chapter. Various aerodynamic measures are then introduced for flutter control, vortex-induced vibration mitigation and cable vibration mitigation. Different mechanical measures – including passive control systems, active control systems, and semi-active control systems – are also addressed for wind-induced vibration control. Damping stay cables of a cable-stayed bridge using adjustable fluid dampers is finally presented as a case study.

12.2 Control Methods for Wind-Induced Vibration

Various possibilities exist to reduce wind-induced vibration of long-span bridges [1,2]. Control methods for reducing wind-induced structural vibration can be divided into three major categories:

1. structural modification;
2. aerodynamic measures; and
3. mechanical measures.

Structural modification is applied to the three important dynamic properties of a bridge structure, i.e. stiffness, damping and mass, through the change of structural materials and/or the rearrangement of structural systems. It can be applied to the entire bridge or bridge components, such as bridge deck, towers, and cables. It is found that an increase in the damping capacity is always beneficial for

mitigating vortex-induced vibration, buffeting-induced vibration and galloping instability. Nevertheless, the effect of additional damping upon flutter-related critical wind speed is low.

In reality, concrete bridges tend to have more damping than steel bridges. Composite steel-concrete bridges also have considerably higher values of damping compared to those of steel bridges. An estimation of the structural damping is crucial but it is the parameter which is least analytically tractable, because its estimation relies totally on limited measured full-scale data.

The common aim of stiffness modification for a long-span cable-supported bridge is to increase the torsional stiffness of the bridge deck so as to improve the bridge stability under wind loading. For example, deep truss girders are more torsionally rigid compared with flat box girders, but they exhibit higher wind forces [3]. To increase the torsional stiffness of a super long-span suspension bridge, cable system modifications may be considered by using more than two main cable systems[4], as shown in Figure 12.1. The deployment of eccentric mass was also applied during construction of the Humber Bridge in England [5], and for the bridge in service [6] if it has relatively close natural frequencies of vertical and torsional motions. The drawback of mass modification is an enlarged dead load and static torsional deformation. To reduce vibration of stay cables in a cable-stayed bridge, cross-ties have been used to increase the in-plane stiffness of stay cables by connecting them together, but this may affect the aesthetic view of stay cables due to a complex cable network and connections.

Aerodynamic measures alter the flow pattern around bridge components (such as bridge deck, towers and cables) to reduce directly the aerodynamic force or moment acting on it by selecting better cross-sectional shapes or by installing some aerodynamic devices on the bridge component. A typical example is to install wedge-shaped fairings to the two longitudinal edges of the bridge deck (see Figure 12.2) to make the bridge deck section streamlined so as to reduce vortex shedding, abate buffeting forces and improve flutter instability [7].

Since wind-bridge interaction is so complicated, no analytical methods are available at present to facilitate the optimal selection of cross-sectional shapes. Wind tunnel tests and computational simulations, as discussed in Chapter 7 and Chapter 8 respectively, have to be used to perform aerodynamic modifications. Although many types of aerodynamic devices, such as vanes and stabilizers, have been found to be effective in the suppression of wind-induced response and the improvement of motion stability, the alterations to the shape of the bridge deck and the substantial nature of the alterations may be impractical and aesthetically undesirable in modern long-span bridges. In particular, the addition of aerodynamic devices to a bridge deck often increases buffeting forces.

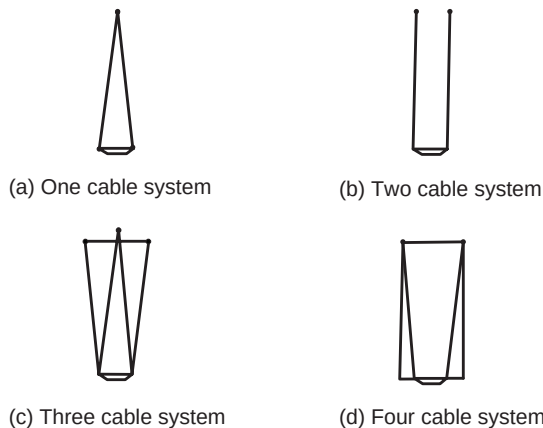


Figure 12.1 Cross-section of alternative suspension cable systems.

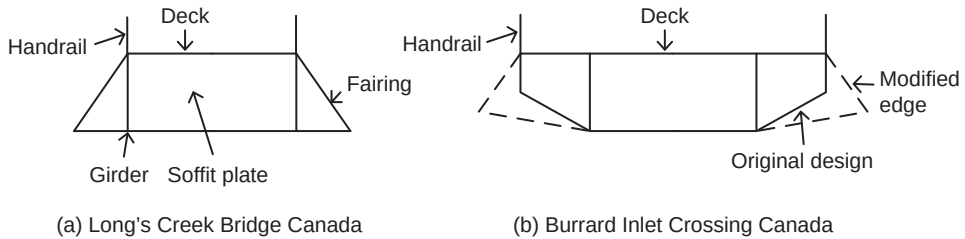


Figure 12.2 Aerodynamic measures to reduce wind-induced vibration.

Mechanical measures, which include passive control technology, active control technology, semi-active technology and hybrid control technology, modify overall structural characteristics, leading to a reduction of wind-induced vibration [8]. They can be applied to the bridge as a whole or to any individual bridge component. Passive control systems operate without requiring an external power source and utilize the motion of a bridge structure to develop control forces. Passive control devices, such as metallic dampers, friction dampers, viscoelastic dampers, tuned mass dampers and tuned liquid column dampers, do not increase vibrational energy and are thus inherently stable [9]. The passive control devices mainly enhance the overall damping capacity of a bridge structure, so as to absorb the energy of wind-induced vibration and thereby reduce it. The basic configurations of a conventional structure and a structure with a passive control system are shown schematically in Figure 12.3a and 12.3b, in which PED stands for passive energy dissipation [10].

Compared with passive control systems, active control systems require external power supply to apply forces to a bridge structure in a prescribed manner. Control forces are developed on the basis of feedback from sensors which measure the excitation and/or the response of the bridge structure. These forces can be used to add or dissipate energy in the bridge structure. An active control system usually consists of:

- (a) sensors installed at suitable locations of the bridge structure to measure either the external excitations, or the structural response quantities, or both;
- (b) devices to process the measured information and to compute the necessary control forces based on a given control algorithm; and
- (c) actuators, usually powered by external energy sources, to produce the required forces.

The basic configuration of a structure with an active control system is shown schematically in Figure 12.3c [10,11]. An active control system mainly changes the overall structural characteristics to reduce wind-induced vibration, but it needs an external power supply and may cause instability of the bridge structure because of many uncertainties involved in the bridge structure, wind loading and control system. Active control systems are also proposed to control some aerodynamic devices to change flow patterns around and reduce wind forces on the bridge deck.

Semi-active control is a compromise between passive and active control systems [12–14]. Semi-active control systems require much less input energy in comparison with the active type, and the input energy is used to modify the damper properties, leading to the optimal control of structure vibration. Control forces are also developed based on feedback from sensors which measure the excitation and/or the response of the structure. Examples include variable orifice dampers, variable friction damper system and magnetorheological (MR) dampers. Figure 12.3d shows the basic configuration of a structure with a semi-active control system schematically.

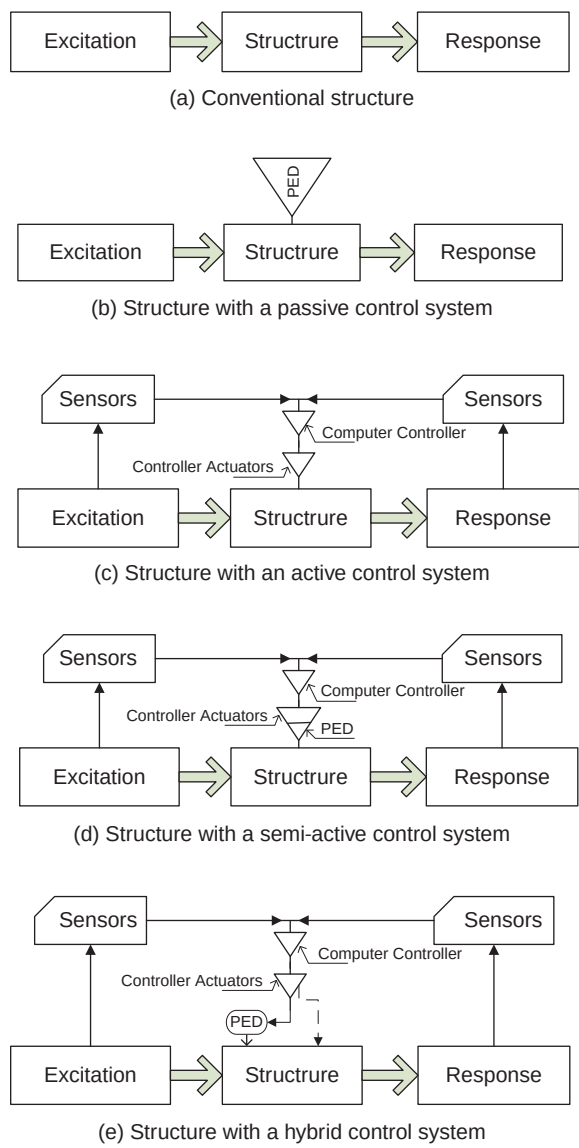


Figure 12.3 Structure with various control systems.

A hybrid system consists of an active system and a passive control system and thus increases the performance and robustness of the control system (see Figure 12.3e). A hybrid mass damper (HMD) is a combination of a passive tuned mass damper and an active control actuator. This has been employed in full-scale building structures, but not yet in bridge structures.

12.3 Aerodynamic Measures for Flutter Control

Flutter instability will lead to the structural collapse of a long-span cable-supported bridge, and hence the critical wind speed associated with flutter instability of the bridge needs to be higher than the maximum design wind speed at the bridge site. When a long-span bridge is predicted to have its intrinsic limit in reaching the required critical wind speed, it is necessary to adopt some aerodynamic or mechanical measures to improve its aerodynamic performance.

12.3.1 Passive Aerodynamic Measures

Aerodynamic forces arise when air flows around bridge deck sections and interacts with them. Passive aerodynamic measures thus aim to directly reduce the aerodynamic forces through improving the configuration of cross-sections of the bridge deck. Passive aerodynamic measures are the most common approach among all the flutter control methods for long-span cable-supported bridges. Some commonly-used measures are introduced below.

Wedge-shaped fairings added at both ends of the bridge deck cross section (see Figure 12.2) can make the cross-section tend to streamline and, hence, improves the air flow pattern around the deck, reduces drag forces and vortex shedding and enhances flutter stability. Wind tunnel studies and flutter analysis have shown that such measures can effectively improve the flutter stability of cable-supported bridges, and the smaller the tip angle of the edge fairings, the greater the improvement of flutter stability [7]. With the same tip angle of the edge fairings, the increase of tip length will enhance the aerodynamic stability of the bridge.

The aerodynamic performance of a bridge deck can be further enhanced by longitudinal open slots in the deck [15–22]. This is a unique feature observed from recently constructed suspension bridges. The adoption of a “slotted” deck solution not only enhances aerodynamic stability, but also prevents the periodic formation and shedding of large vortices in the wake of the bridge deck.

The aerodynamic performance of the slotted box girder can be further improved if it is used together with some aerodynamic devices. Ueda *et al.* [23] investigated the flutter stability of a super-long-span suspension bridge with a central span length of 2500 m by considering three types of central slots: central slots only, central slots with central stabilizer, central slots with both central stabilizer and horizontal stabilizer (see Figure 12.4).

The wind tunnel test results showed that the critical flutter wind speed increased by about 35% and reached 62 m/sec after the central stabilizer was added to the central slots. If a horizontal stabilizer was further added to the central slots with the central stabilizer, the critical flutter wind speed was improved again by 33% and reached 82.2 m/sec. The aerodynamic advantages of this solution have also been explored during the design of the proposed 3300 m long suspension bridge for the crossing of the Messina Strait [24], and the relevant wind tunnel studies were performed rigorously by Diana [25].

Other alternative passive aerodynamic measures involve modifying wind loading acting on a bridge by adding aerodynamic appendages to the deck section [26]. The idea is to locate a winglet above (or below) the bridge deck, running parallel to the bridge axis, far enough from the deck that it lies in the undisturbed flow field. The presence of the winglets modifies the aeroelastic loads acting on the bridge and increases the flutter associated with the critical wind speed. Arco and Aparicio [27] investigated four types of aerodynamic appendages added to the bridge cross-section according to the position of the winglets, as shown in Figure 12.5.

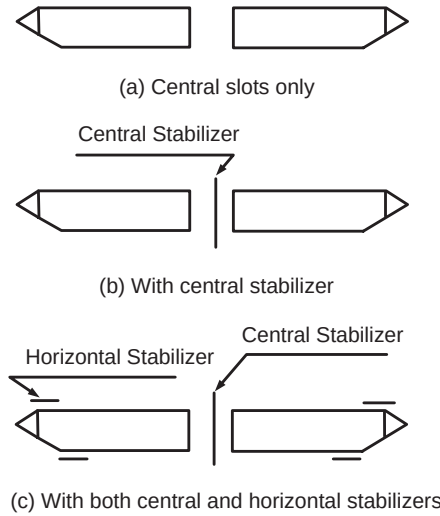


Figure 12.4 Slotted box girder with central stabilizer and horizontal stabilizer.

In Case A, a pair of winglets are symmetrically fixed to the bridge cross-section. In Case B, a pair of winglets are fixed to the hangers. In Case C, only one winglet is fixed to the bridge cross-section in the leeward position. In Case D, a pair of winglets are fixed to both the windward and leeward positions. These results demonstrated that, compared with Cases A and B, Case C gave the best aerodynamic performance and the highest critical wind speed. When this finding was applied to the Great Belt Bridge, using aerodynamic appendages arranged like Case C, it was found that the critical wind speed was clearly enhanced. The increase in critical wind speed in Case C accounted to more than 35%, compared with the normal case without any appendages. The increase of critical wind speed in Case B was somewhat modest, at 21%, and was even less (11%) in Case A. For Case D, care must be given to the selection of the phase angle, for a wrong selection of the angle may cause the instability of the bridge structure.

All of these passive aerodynamic measures are viewed as economical and reliable methods that do not require special routine maintenance. They have been validated through wind tunnel tests, computational simulations, and field measurements.

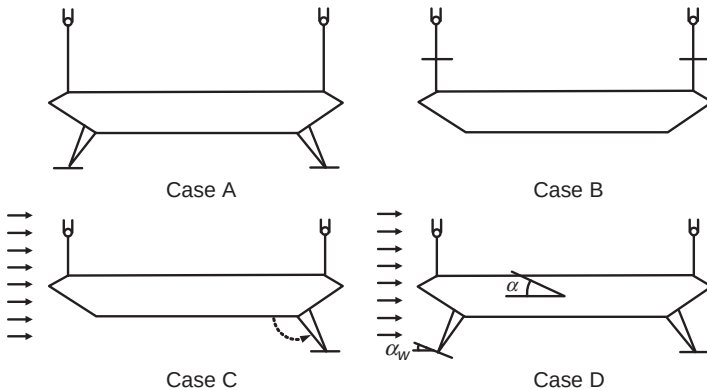


Figure 12.5 Different arrangements of aerodynamic appendages (from [27]).

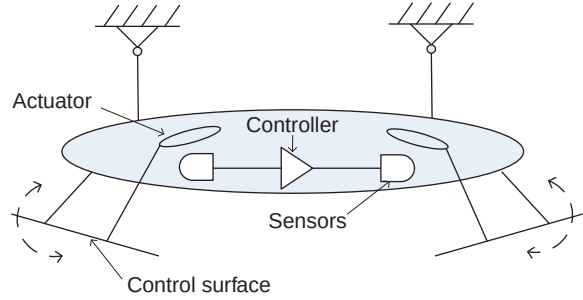


Figure 12.6 Active aerodynamic control of bridge deck flutter by control surface.

12.3.2 Active Aerodynamic Control

Active aerodynamic control methods for long-span bridges are inspired by those employed in the aerospace industry for aircraft control. Ostenfeld and Larsen [3] presented an active aerodynamic control system by using actively controlled surfaces attached beneath the two edges of the bridge deck cross-section (see Figure 12.6).

The rotations of the surfaces are actively controlled according to a feedback algorithm in such a way that the aerodynamic forces acting on the bridge deck provide a stabilizing action. These stabilizing aerodynamic forces are not produced directly by the actuators in this method, and they actually result from the change of air flow around the bridge deck via the rotation of the surfaces controlled by the actuators.

Kobayashi and Nagaoka conducted wind tunnel experiments on the sectional model of a bridge and obtained an increase of flutter critical wind speed of factor 2 [28]. The control algorithm used in the experiments was based on the principle that the rotation of the control surfaces is proportional to the rotation of the bridge deck. Wilde and Fujino [29] carried out theoretical analysis of such a control surface system. They applied a rational function approximation to model unsteady aerodynamics and, under the assumption of no flow interaction between the control surfaces and the deck, they derived a time domain equation of motion for the control system. Accuracy of the approximation was achieved by multilevel linear and non-linear optimization. Their numerical simulation results show that the aerodynamic control of flutter by actively controlled surfaces can stabilize the bridge at any wind speed. For the control to be effective, it is necessary to locate control surfaces outside the turbulent boundary layer and as far away from the local flow pattern around the deck as is practically possible.

Active aerodynamic control may also be achieved by additional flaps attached directly to the edges of the bridge deck [30]. In this system, the flow pattern around the deck is affected by the motion of the flaps (see Figure 12.7). Therefore, the stabilizing action not only comes from the aerodynamic forces generated on the flaps, but also from modification of the aerodynamic forces induced on the bridge deck.

Wilde *et al.* [31] referred to this control system as the “deck-flaps system” and presented the modeling procedure and the control algorithm for controlling flutter of long-span bridges. Three optimal configurations of the deck-flaps system, with different flap hinge locations, were proposed and investigated. For each hinge location, four flap sizes were considered: 1.5 m, 3.0 m, 4.5 m and 6.0 m. All of

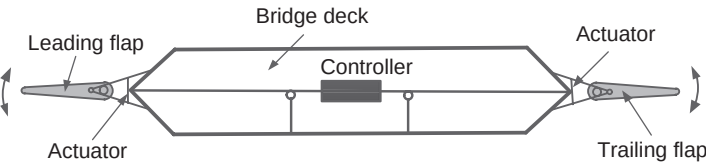


Figure 12.7 Cross-section of bridge deck with active flaps for flutter control.

these considered configurations of the control system could stabilize the bridge against flutter and divergence in the selected wind speed range. The best performance was obtained by the system with the flaps located on the edges of the deck and the hinges located in the middle of the flaps.

Omenzetter *et al.* [32–35], on the other hand, proposed similar control systems but of the passive type, consisting of auxiliary flaps attached directly to the bridge deck, as shown in Figure 12.8.

When the deck undergoes pitching motion or relative horizontal motion with respect to the main cables, the flap rotation is governed by additional cables spanned between the flaps and an auxiliary transverse beam supported by the main cables of the bridge. Since the cables can only pull the flaps,

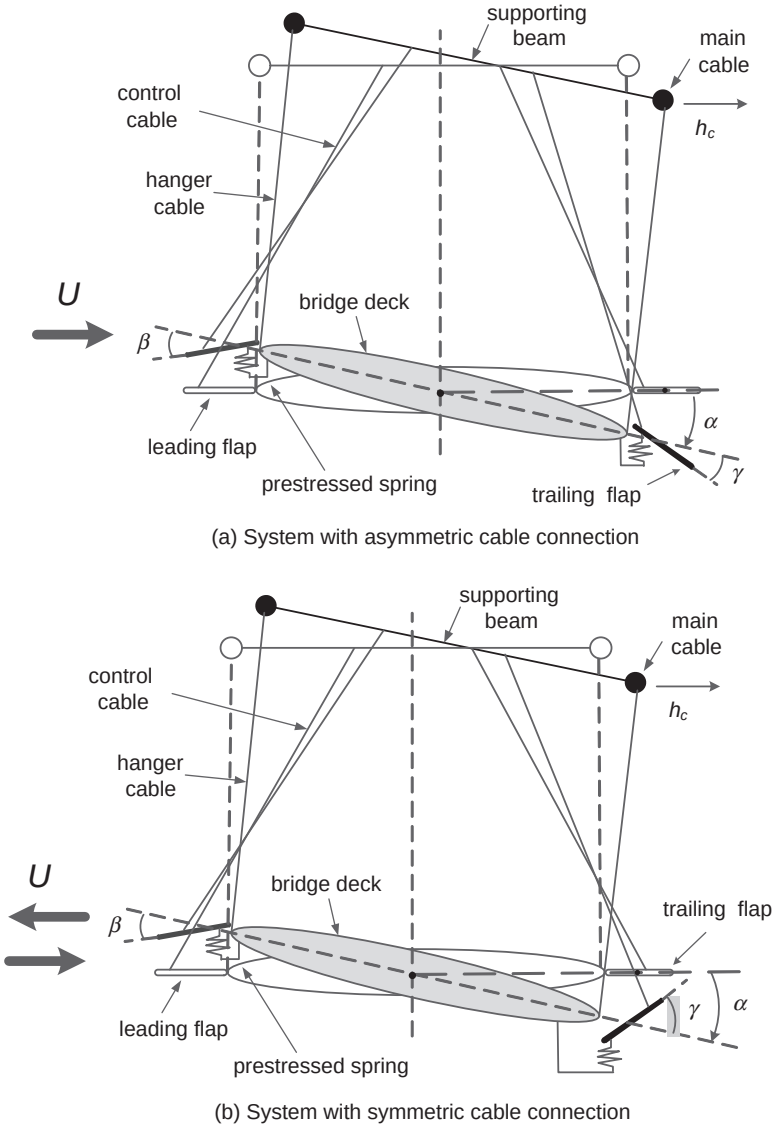


Figure 12.8 Aerodynamic control of bridge deck flutter with passive flaps (Figure courtesy of Prof. Yozo Fujino).

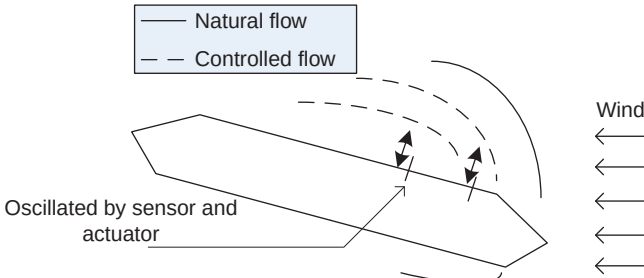


Figure 12.9 Active aerodynamic control by changing air flow.

but not push them, additional pre-stressed springs are used to reverse the motion of the control surfaces. The control system shown in Figure 12.8a is an asymmetric cable connecting system, which can work properly for wind coming from only one direction and requires alteration of its configuration as wind direction changes. The stabilizing action of the system shown in Figure 12.8b comes from a symmetric cable connection system, which is, in contrast, independent of wind direction.

For these two systems, Omenzetter *et al.* [32–35] derived the time-domain mathematical model of self-excited forces on the sectional model of the deck-flaps system and conducted numerical simulations of a suspension bridge using a three-dimensional FEM model.

Although both systems employ the same idea of flutter control, their dynamic characteristics and performance differ significantly. The system with asymmetric cable connection offers high increase in critical wind speed. The action of control flaps strongly modifies the properties of the flutter mode and its stiffness and damping are markedly increased. However, this system requires changes in its configuration as wind direction changes. To achieve its best performance, it requires large flaps and significant stiffness of the supporting system.

The improvement in critical wind speed for the system with symmetric cable connection is limited. Properties of the flutter mode are not strongly altered and the degree of stability is lower. The gust response of the control flaps is large. Nevertheless, this system can work properly for the wind coming from any direction. It uses flaps of small size, and the stiffness of the supporting system is much less compared to the system with asymmetric cable connection.

There was an active aerodynamic control idea for flutter in aeronautics that utilized a fence-type spoiler to change airflow around the airfoil surface, as shown in Figure 12.9 [36]. Kwon *et al.* [36] applied similar idea to a bridge deck and designed a new aerodynamic controller to control plates properly according to the motion of the bridge deck without requiring external power supply.

Figure 12.10 schematically shows the mechanism of the proposed aerodynamic control method. The key idea of this system is to properly activate the control plates according to the bridge deck motion. A constant phase angle between the control plate and deck motion should be kept for successful control.

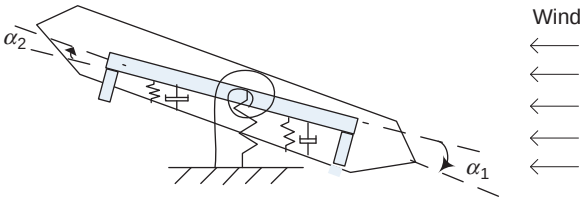


Figure 12.10 Control mechanism of active aerodynamic control system.

There are two slits on the bottom flange of the deck, and the control plates oscillate through these slits. When the bridge deck maintains its horizontal state, the control plates do not move and the deck keeps its equilibrium position. However, when the deck is oscillated by the wind, the control plates move up or down, according to the motion of the decks, and thus the air flow is changed.

A seesaw-like rod inside the deck activates the control plates. Its mechanism is similar to a tuned mass damper (TMD). The phase angle between the deck and control plate can be adjusted by the mass and stiffness of the rod. Because the inertial force of a TMD is used for moving control plates for aerodynamic control and not for direct vibration control, the mass ratio of this system can be greatly reduced in comparison with a traditional TMD for structural control. This is one of the great benefits of the proposed system.

A spring-supported section model test was conducted in order to verify the proposed control method. The wind tunnel test results showed that the increasing ratio of the critical flutter wind speed was 64.6. For studying the applicability of the proposed system to a long-span bridge, a long-span suspension bridge with a central span length of 3000 m was used as a numerical example. It showed that the flutter onset velocity increased by 51% when 60% of the center span equipped the controller.

12.4 Aerodynamic Measures for Vortex-Induced Vibration Control

Since the selection of a bridge deck configuration depends on many factors, such as structural and economical advantages, the basic deck shape does not necessarily have optimal aerodynamic efficiency. As a consequence, long-span bridges may be subjected to vortex-induced vibration. However, compared with the flutter control of long-span bridges, the studies on vortex-induced vibration control are relatively weak. The following will give a brief introduction on aerodynamic measures for vortex-induced vibration control of long-span cable-supported bridges.

During the final phase of deck erection and road surfacing works of the Storebælt suspension bridge in Denmark in 1998, vertical deck oscillations of low frequency were observed [37]. The results from both field measurements and wind tunnel tests indicated that the observed oscillations were due to periodic von Karman type vortex shedding in the wake of the mono box girder of the bridge. With the aid of wind tunnel tests, Larsen *et al.* [37] then designed and installed guide vanes to the main suspension girder for suppression of vortex-induced vibration of the bridge (see Figure 12.11).

The guide vanes were composed of curved steel plates running along the bottom/lower side panel joints and were tested on a 1: 60 scale section model of the bridge girder. This guide vane design was proved very efficient in eliminating vortex-induced oscillations of the bridge by both field measurements and wind tunnel tests. In addition, the guide vanes improved the flutter stability of the bridge deck marginally, with the critical wind speed for onset of flutter rising from 69 m/sec to 73 m/sec.

The Stonecutters cable-stayed bridge in Hong Kong has a main span of 1018 m, and the bridge deck is composed of two wedge-shaped box girders with an open gap between them (see Figure 12.12). Vortex shedding and lock-in with the lowest vertical mode of the bridge was expected at low wind

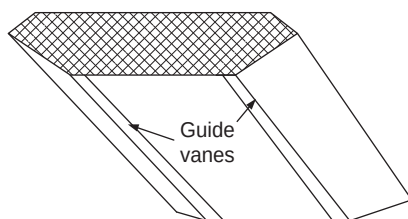


Figure 12.11 Guide vanes installed in the Storebælt suspension bridge.

The Xihoumen Bridge in China is a long-span suspension bridge with a 1650 m main span. The bridge deck is a twin box girder. Ge *et al.* [39] carried out both conventional sectional model tests and large-scale sectional model tests of the bridge and found vortex-induced vibration with large amplitude. Several aerodynamic measures, including guide vane, grid plate and adjustable wind barrier, were proposed and tested to find the most effective and feasible measure that could be applied to the real bridge. Later, vertical vortex-induced vibration did occur in the prototype bridge [40]; an adjustable wind barrier was then selected and installed on the girder deck to suppress the vortex-induced vibration of the bridge.

12.5 Aerodynamic Measures for Rain-Wind-Induced Cable Vibration Control

With widespread popularity of cable-stayed bridges around the world, longer spans are being constructed by employing increasingly longer stay cables. These cables are laterally flexible structural members with very low fundamental frequency and very little inherent damping and, for this reason, they have been known to be susceptible to wind-induced vibration, as well as rain-wind-induced vibration [41,42]. Recognition of this susceptibility of stay cables leads to the use of some aerodynamic measures. The following is a brief introduction to the aerodynamic measures used for rain-wind-induced vibration control of stay cables on the Stonecutters Bridge. The aerodynamic measures considered were helical fillets and dimples added on the cable surface. Wind tunnel investigation was carried out to determine drag coefficients of the 1: 1 scale section models of stay cables manufactured with a diameter of 140 mm, 160 mm and 180 mm, which are representative of medium, long and very long prototype cables of the Stonecutters Bridge [43]. Rain-wind tunnel investigation was also conducted to assess the efficiency of helical fillets and dimples for mitigating rain-wind-induced cable vibration [44].

12.5.1 Wind Tunnel Investigation and Cable Drag Coefficients

Cable section models were designed and made of 1: 1 scale stainless steel cylinders covered with high density polyethylene tube, which is the same material as used in the prototype stay cables, to ensure Reynolds number similarity and the same surface conditions for cable model and prototype. The wind tunnel tests were conducted in the high-speed working section of the CLP Wind Tunnel at the Hong Kong University of Science and Technology. Figure 12.14 shows some of the cable section models used in the wind tunnel investigation, including cables with smooth surface, cables with helical fillets and cables with dimples. Detailed information on the wind tunnel and the test apparatus can be found in Section 7.10 in Chapter 7.

Helical fillets of a rectangular cross-section were stuck on the smooth surface of the cable model, with an effective step of 0.3 m, to simulate the prototype cable with helical fillets. The helical fillet was made of light wood with a square cross-section of 2×2 mm. A cable model of dimpled surface was manufactured of a diameter of 140 mm. The shape and density of the dimples on the surface of the cable model are shown in Figure 12.15.

The lengths of major axis and minor axis of the elliptic dimple were 8.5 mm and 6.5 mm, respectively. The periphery of the elliptic dimples was about 0.5 mm above the cable surface, while the central part of the dimples was pressed into the cable by about 1.5 mm. Dimples of elliptical shape and 2 mm depth were made on the surface of the cable model in the pattern shown in Figure 12.15. The density ratio of the surface area of the dimples to the surface area of the cable was approximately 0.14.

The drag coefficients of stay cables of different configurations were investigated through wind tunnel tests with respect to cable diameter, inclination and yaw angle. Figure 12.16 exhibits the test rig used and the experimental set-up. The aerodynamic force coefficients of the stay cables of smooth surface were compared with those of the stay cables of either helical fillets or dimples, so that the difference in static wind load on the stay cables due to surface change could be estimated. The major results obtained



Figure 12.14 Selected cable section models used in wind tunnel investigation.

from the wind tunnel investigation could be summarized in the following, and the full results can be found in the reference [43].

The drag coefficients of the stay cables with a smooth surface were larger when wind was perpendicular to the bridge deck axis than when the wind was along the bridge deck axis. When the wind was perpendicular to the bridge deck axis, the drag coefficients of the stay cables decreased, with increasing mean wind speed ranging from 10–20 m/sec. The drag coefficients of the stay cables with the helical fillets were only slightly larger than those of the same cables with smooth surface when the wind was perpendicular to the bridge deck axis. When the wind was along the bridge deck axis, the drag coefficients of the stay cables with helical fillets may not be larger than those of the same cables with smooth surface. The drag coefficients of the cable with dimpled surface were smaller than those of the cable with smooth surface in all the wind directions concerned. The drag coefficient of the dimpled cable

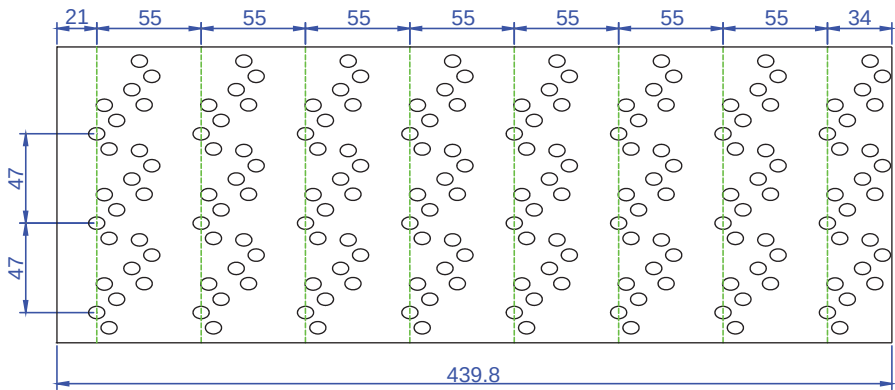


Figure 12.15 Dimple pattern on the cable surface (unit in mm).

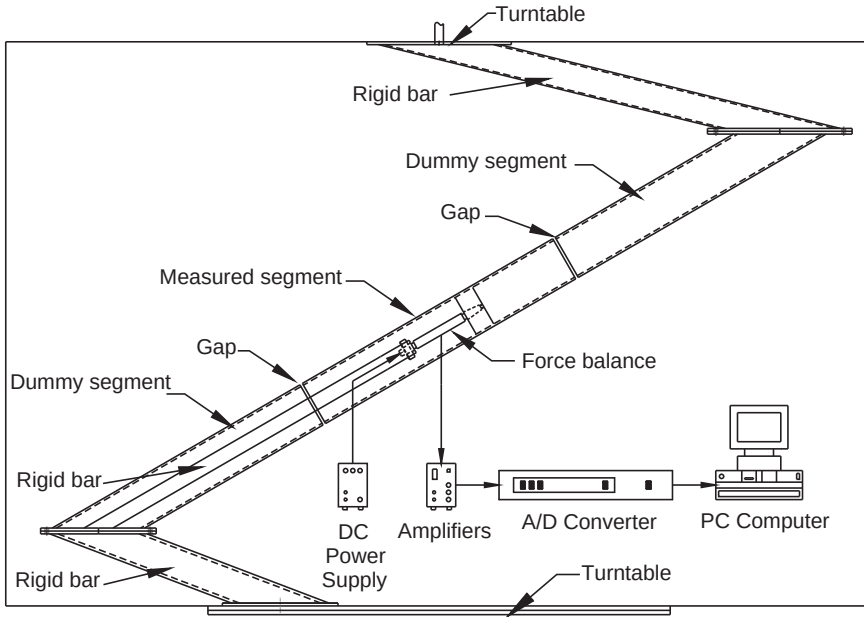


Figure 12.16 Wind tunnel test arrangement for inclined cables (from [63]) (Reproduced with permission from Multi-Science Publishing Co Ltd).

was also much less influenced by wind speed, compared with the smooth cable. In summary, static wind load on the cable with dimpled surface is the smallest among the three types of cable surface.

12.5.2 Rain-Wind Tunnel Investigation of Stay Cables of Different Surfaces

Cable section models were also designed and made of 1: 1 scale stainless steel cylinders covered with high density polyethylene tube, which is the same material as used in the prototype stay cables. The diameters of two cable models tested were 139 mm (140) and 169 mm (170) respectively. Three cable surfaces were considered: cables with smooth surface, cables with helical fillets, and cables with dimples.

Figure 12.17 shows the schematic diagram of the test arrangement. The natural frequency was 0.63 Hz for the cable model of 170 mm diameter and 0.91 Hz for the cable model of 140 mm diameter. With the aid of an oil damper, the damping ratio of the test cable model varied within a range from 0.1% to 1%. The Scruton number of the test cable model also varied within a range from 0.9 to 8. The inclination of the cable model denoted by angle α was set as 21° and 36° respectively, while the yaw angle of incident wind designated by angle β varied from 0° to 65° . The rain-wind tunnel tests were conducted by the Southwest Jiaotong University in China. The major results obtained from the investigation could be summarized in the following, and the full results can be found in the reference [44].

For the 170 mm diameter cable model, rain-wind-induced vibrations were clearly observed for the yaw angle of incident wind from 30° to 55° . The strongest cable vibration occurred at $\beta = 30^\circ$ and a wind speed of 12 m/sec. The maximum amplitude of cable vibration was up to 440 mm, about 2.6 times the cable diameter.

For the 140 mm diameter cable model, rain-wind-induced vibrations were found for all the tested yaw angles. The strongest cable vibration occurred at $\beta = 35^\circ$ and a wind speed of 10.5 m/sec. The maximum amplitude of cable vibration was more than 138 mm, almost equal to the cable diameter.

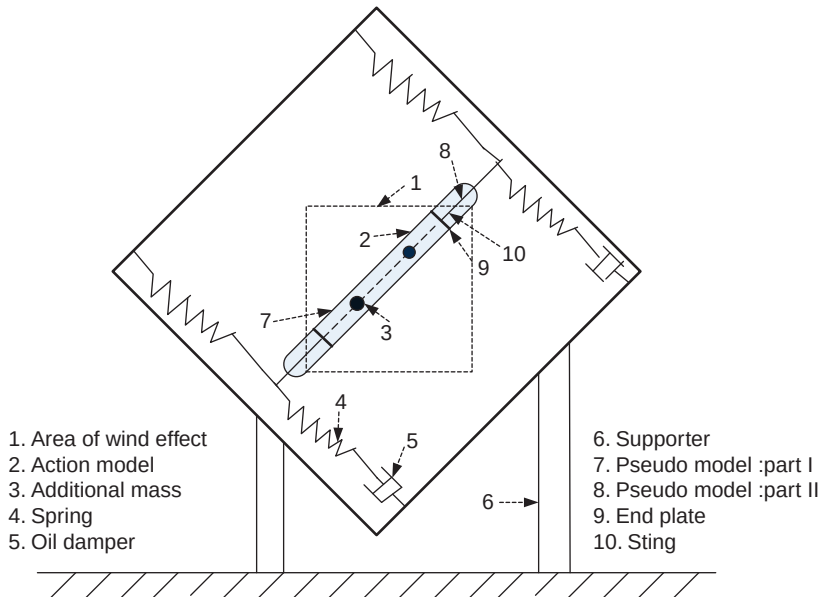


Figure 12.17 Rain-wind tunnel test arrangement for inclined cables.

As for the effects of rain intensity, it was found that the strongest cable vibration emerged at a rain intensity of 75 mm/h for the cable model of 170 mm diameter. The maximum amplitude of cable vibration was about 482 mm, about 2.85 times the cable diameter. For the cable model of 140 mm diameter, the strongest vibration emerged at a rain intensity of 85 mm/h. The maximum amplitude of cable vibration was about 153 mm, which was 1.1 times the cable diameter.

For the cable model with either helical fillets or dimples, no significant cable vibration was observed for most of the test cases. The vibration amplitude of the cable model with helical fillet increased slightly with the increasing height of the fillet. Relatively strong vibration was only observed for the cable model of 170 mm diameter with dimple at $\beta = 40^\circ$ and a rain intensity of 80 mm/h. The maximum vibration amplitude was about 0.95 times the diameter of the cable model.

It could be concluded that the helical fillets could reduce rain-wind-induced cable vibration significantly but amplify static wind load. The dimples could reduce both rain-wind-induced cable vibration and static wind load, but rain-wind-induced cable vibration might still emerge for particular wind directions and cable diameters.

12.6 Mechanical Measures for Vortex-Induced Vibration Control

Sometimes, particularly after a bridge is completed, it is difficult to find aerodynamic measures that are effective for vortex-induced vibration control. When satisfactory aerodynamic solutions cannot be found, the control of vortex-induced vibration relies on mechanical measures. Most popular mechanical measures for vortex-induced vibration control are tuned mass dampers (TMD). A TMD is a device consisting of a mass attached to a structure in such a way that it oscillates at the same frequency of the bridge structure, but with a phase shift. The mass is attached to the structure via a spring-dashpot system, and the energy is dissipated by the dashpot as relative motion develops between the mass and the structure. The schematic diagram of a TMD attached to a flexible beam is shown in Figure 12.18.

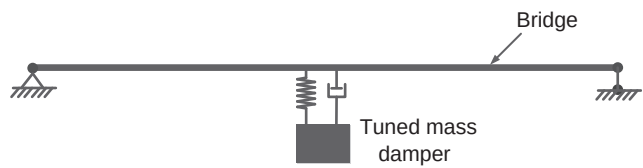
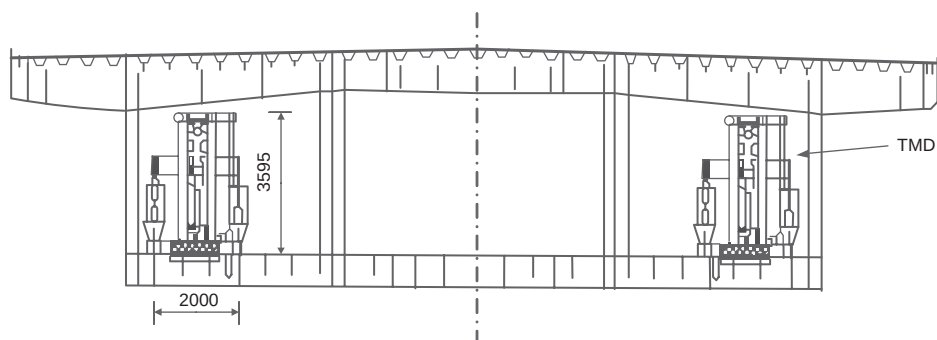
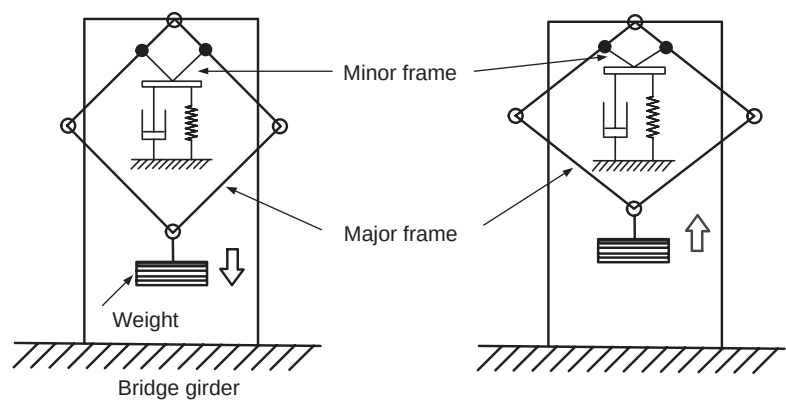


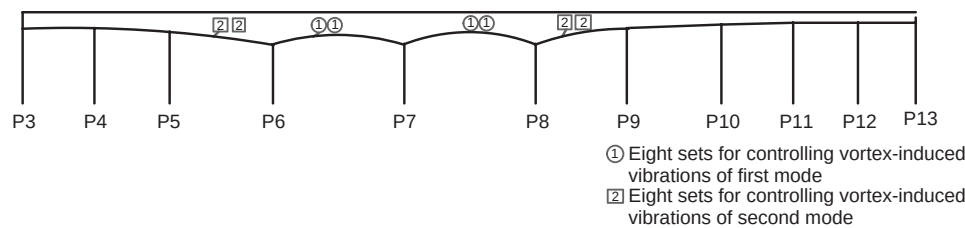
Figure 12.18 Schematic diagram of a TMD attached to a bridge.



(a) Layout of TMDs (cross-sectional view)



(b) Mechanism of TMD



(c) Layout of TMDs (longitudinal view)

Figure 12.19 Tuned mass dampers and their locations of Trans-Tokyo Crossing Bridge (Figure courtesy of Prof. Yozo Fujino).

The Trans-Tokyo Bay Crossing Bridge in Japan, completed in 1997, is a ten-span continuous steel box girder bridge with a total length of 1630 m [45]. Although this bridge is not a cable-supported bridge, the experience of vortex-induced vibration control obtained from the bridge is invaluable for long-span cable-supported bridges. In the design phase of the bridge, various wind tunnel tests were carried out on two-dimensional sectional models as well as a complete three-dimensional model [46]. These tests revealed that vortex-induced vertical flexural vibration would develop in the bridge under the wind normal to the bridge axis. A variety of aerodynamic mitigation strategies were tested, but complete suppression of vibration was not achieved. As a result, consideration was given to adding TMDs to the bridge if excessive vibration did happen after completion of the bridge.

Significant vortex-induced vibration was, indeed, observed after the girder erection was completed in 1994. The vortex-induced first-mode of vibration peaked at a wind speed of around 16–17 m/sec with maximum amplitude over 50 cm [45]. It was decided to install 16 TMDs in the girders to control its first and second modes of vibration. Vibration in the higher modes was controlled aerodynamically by attaching vertical plates to the posts of crash barriers in the girders.

The TMDs for the bridge were installed inside the box girder, as shown in Figure 12.19a. The structure of a TMD consisted of a major frame and a minor frame in the shape of a pantograph (see Figure 12.19b). The weight of the TMD was mounted on the lower end of the major frame via coil springs and oil dampers, and the TMD moved up and down with girder vibration. The use of two frames made it possible to reduce the TMDs' size considerably. Eight TMDs were installed for the first mode and eight for the second mode, as shown in Figure 12.19c.

A two degrees of freedom model was adopted in the determination of the parameters of the TMDs, and those for the first mode were designed independently of those for the second mode. The natural frequency of the TMDs was 0.33 Hz for the first mode and 0.47 for the second mode, and the mass ratio was 1.16% for both modes of vibration. The logarithmic decrement damping of the bridge-TMD system was targeted to be 0.22 for both modes of vibration. The maximum stroke of the TMD was ± 600 mm for the first mode and ± 800 mm for the second mode. The field measurements were made before and after installation of the TMDs, to access their performance. The measurement results showed that the performance of the TMDs was satisfactory: under the almost same wind conditions, the amplitude of girder vibration without TMDs was over 40 cm, but the amplitude of girder vibration with the TMDs was just 5–6 cm.

12.7 Mechanical Measures for Flutter Control

A number of studies have been conducted by utilizing auxiliary damping devices for controlling bridge flutter, although most of them have not yet been implemented in real long-span cable-supported bridges. These devices include passive control systems, active control systems and semi-active control systems.

12.7.1 Passive Control Systems for Flutter Control

Various passive control systems, such as TMDs, tuned liquid dampers (TLD), and tuned liquid column dampers (TLCD) have been proposed and investigated for controlling bridge flutter. Nobuto *et al.* [47] carried out both numerical analyses and wind tunnel tests of a bridge section model to investigate the effectiveness of TMDs for flutter control. Their results showed that the critical flutter wind speed was increased by about 14%. However, the performance of the TMD system was very sensitive to the tuning condition. Gu *et al.* [48] also conducted both numerical analyses and wind tunnel tests of a box section model to confirm the performance of TMDs for controlling flutter. Two TMDs were placed on the leading and trailing edges of the deck, and their mechanical model is shown in Figure 12.20.

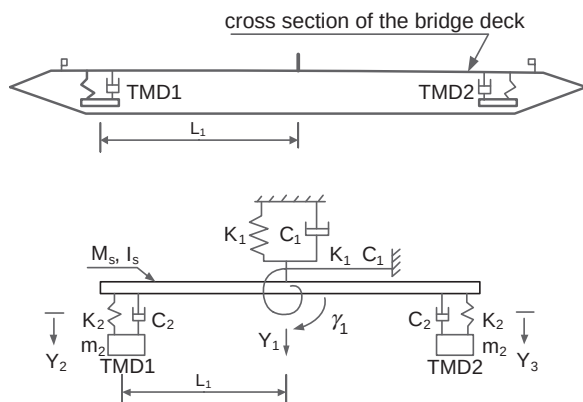


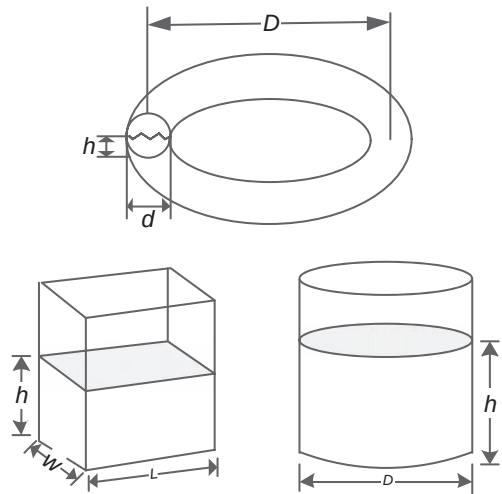
Figure 12.20 Mechanical model of TMDs and bridge deck for flutter control (from [48]).

Their results showed that TMDs could increase the critical flutter wind speed significantly. The TMDs with more than 5.6% inertia mass moment ratio could increase the critical flutter wind speed of the Tiger-Gate Bridge section model with wind screens by more than 40%. Another TMD study was carried out by Pourzeynali and Datta [49], in which a combined vertical and torsional TMD system was applied to increase the critical flutter wind speed. The proposed system had two degrees of freedom, tuned close to the frequencies corresponding to the vertical and torsional symmetric modes of the bridge for coupled flutter control. The maximum improvement of the critical flutter wind speed was about 2.03 times the uncontrolled critical wind speed for a 5% TMD damping ratio.

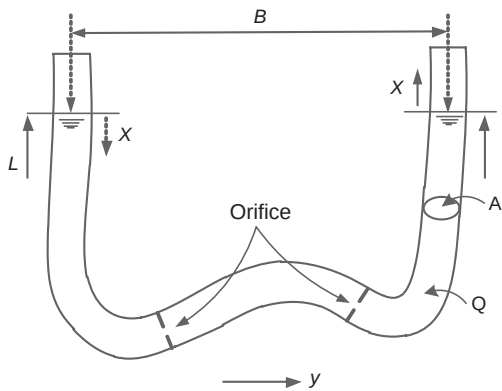
However, all of the above studies indicated that the performance of a TMD is very sensitive to frequency tuning: a slight shift of the frequency tuning may lead to a significant deterioration of control performance of the TMD. Kwon and Park [50] therefore proposed to use multiple-tuned mass dampers (MTMD) to overcome the problem. They investigated two types of MTMD – some with frequencies that were equally spaced and some that were not equally spaced. They found that irregular multiple-tuned mass dampers that have unequal frequency interval and different damping ratios offered more flexible designs and provided better performance for controlling bridge flutter.

While the effectiveness of TMDs in controlling bridge flutter has been demonstrated in these studies for specific bridges, their limitations and the dependence of control performance on the dynamic and aerodynamic characteristics of bridge structures were addressed by Chen and Kareem [51]. Their study showed that the performance of TMDs strongly depends on the bridge's aerodynamic characteristics. For a hard-type flutter characterized by negative damping that grows rapidly with increasing wind speed beyond the flutter onset, the influence of structural modal damping on the critical flutter speed was insignificant. Accordingly, the effectiveness of TMDs in controlling this type of flutter was rather marginal. However, for a soft-type flutter, in which the negative damping of bridges grows slowly with increasing wind speed, addition of auxiliary damping may result in significantly higher critical flutter wind speed. For this type of flutter, a reliable estimate of structural modal damping was critical for accurate estimation, and auxiliary dampers such as TMDs would be relatively effective in controlling such type of flutter.

Another class of passive control systems is TLD, in which liquids are used to provide all of necessary characteristics of the secondary system (see Figure 12.21a). In this system, the liquid not only supplies the required secondary mass but also damping through viscous action, primarily in the boundary layers. Meanwhile, the gravity force of the liquid provides the necessary restoring mechanism. As a result, the secondary system has characteristic frequencies that can be used for optimal performance, similar to TMDs, for flutter control.



(a) Tuned liquid damper



(b) Tuned liquid column damper

Figure 12.21 Tuned liquid damper and tuned liquid column damper.

TLCD for controlling torsional flutter and buffeting of a long-span bridge were also investigated by Xue *et al.* [52], experimentally and numerically. The TLCD simply consists of a U-shaped container filled with water, and structural vibration energy is dissipated as the water passes through an orifice with inherent head-loss characteristics (see Figure 12.21b). For controlling torsional flutter of a bridge deck, Figure 12.22 displays a schematic diagram of the experimental setup and instrumentation, in which a TLCD model is installed inside a bridge deck unit model [52].

The bridge deck unit model is supported by a pivot at the middle point of its horizontal frame so that the bridge deck could rotate around the pivot. Two springs are installed at the two ends of the horizontal frame to provide the deck with torsional stiffness. The test platform is rigidly connected to the horizontal frame, using four vertical steel members. The whole deck unit could thus be seen as a single degree of freedom system rotating around the pivot. The container of a TLCD is fixed on the test platform and

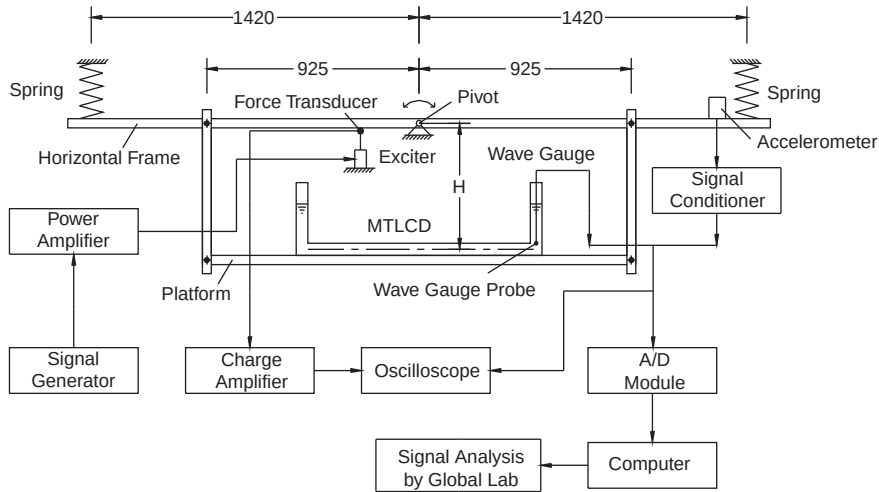


Figure 12.22 Experimental set-up and instrumentation of TLCD for flutter control.

the weight of the container without water is regarded as part of the deck weight. The coupled equations of motion of the bridge girder equipped with a TLCD for torsional vibration control can be expressed as:

$$(I_s + I_d)\ddot{\theta} + \rho AB \left(H + \frac{L-B}{2} \right) \ddot{W} + C_s \dot{\theta} + (K_s + \rho ALgH)\theta + \rho AgBW = M_s \quad (12.1a)$$

$$\rho AL \ddot{W} + \frac{\rho A}{2} \delta |\dot{W}| \dot{W} + 2\rho AgW + \rho AB \left(H + \frac{L-B}{2} \right) \ddot{\theta} + \rho AgB\theta = 0 \quad (12.1b)$$

with the condition:

$$W \leq \frac{L-B}{2} - \frac{d}{2} \quad (12.1c)$$

where:

- I_s is the mass moment of inertia of the bridge girder with respect to the pivot;
- I_d is the mass moment of inertia of water column of the TLCD with respect to the pivot;
- ρ is the density of water;
- A is the cross-sectional area of water column of the TLCD;
- B is the horizontal width of the water column of the TLCD;
- H is the distance from the centre line of the bottom tube of the TLCD to the pivot;
- L is the total length of water column;
- C_s is the damping coefficient of the structure;
- K_s is the torsional stiffness of the structure;
- g is the acceleration of gravity;
- δ is the head loss coefficient of the TLCD governed by the opening ratio of orifice;
- θ is the torsional displacement of the structure;
- W is the relative motion of water to the container;
- d is the thickness of water column;
- M_s is the external moment excitation.

The experimental and numerical results show that TLCD is also an effective device for increasing critical wind speed and for reducing buffeting response of a long-span suspension bridge [52].

12.7.2 Active Control Systems for Flutter Control

With increasing span of cable-supported bridges, a few studies have been made on active and semi-active control systems for flutter control. One of the active control systems being considered is active mass dampers (AMD), which change the dynamic properties of the bridge structure to enhance flutter stability in terms of active control forces or moments generated by the actuators that are, in turn, regulated by control algorithms according to the feedback from the structural responses and/or external excitations. AMDs can be assembled and operate completely inside the bridge girder.

Two types of AMDs were investigated by Korlin and Starossek [53]: the rotational mass damper (RMD) and the movable eccentric mass damper (MEMD). The equations of motion were established on the basis of a two degrees of freedom model of the bridge structure.

Figure 12.23a shows the two degrees of freedom structure with a RMD: the vertical displacement and the rotational displacement. An additional rotating mass was implemented in the center of the bridge girder for active control. The control variable was the rotational acceleration of damper mass. Figure 12.23b displays the two degrees of freedom model of the bridge structure with MEMD. In this case, the active control was exerted by a moment due to gravity force of the additional mass. The eccentricity was the control variable. An optimal linear static feedback controller was used for flutter suppression in both the analysis and the experiment. The motion-induced aerodynamic forces considered in the analysis were obtained experimentally in terms of frequency dependent flutter derivatives. The flutter analysis of the uncontrolled and the controlled bridge structure was carried out using the Hurwitz criterion.

Various control scenarios under different flow regimes were simulated, and the analytical and experimental results were found in good agreement. A deterioration of control performance and flutter stability was found as the effect of saturations in the control. The large energy demand of the proposed AMDs in combination with linear control was a potential challenging issue. Alternative devices and/or control laws should be a topic of further research.

Achkire *et al.* [54] presented another active control strategy for flutter control of cable-stayed bridges. As shown in Figure 12.24, the flutter control of the bridge is realized through active tendon control of stay cables. The control law adopted was decentralized integral force feedback, which guaranteed control stability. The efficiency of the active control strategy was demonstrated by experiments on small laboratory models using piezoelectric actuators, which laid on a foundation for further investigation.

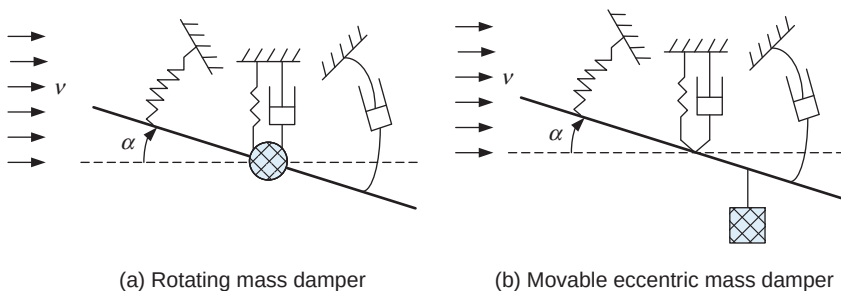


Figure 12.23 Active mass dampers for flutter control of bridge deck.

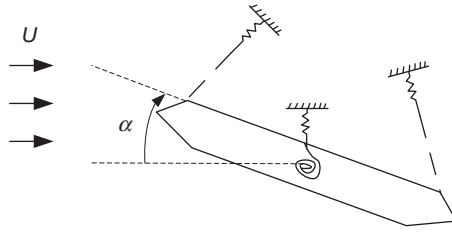


Figure 12.24 Active tendons for flutter control of bridge dec.

12.7.3 Semi-Active Control Systems for Flutter Control

A combined vertical and torsional semi-active TMD (STMD) system was proposed by Pourzeynali and Datta for flutter control of long-span suspension bridges [55]. The semi-active TMDs had adjustable damping which could be actively controlled. A fuzzy logic feedback control algorithm was used to obtain the optimal damping of TMDs for optimal reduction of the bridge response at the flutter wind speed. The numerical results showed the superior performance of semi-active control over passive control.

For the uncontrolled case, the critical flutter wind speed was about 55.52 m/sec. With a passive TMD system having a 20% damping ratio, the critical flutter wind speed increased to 98 m/sec. With nearly the same value of the maximum variable damping (21.6%) of the STMD system, the bridge underwent a decaying oscillation at a wind speed of 110 m/sec. The maximum torsional oscillation was reduced from 0.02 rad to 0.0063 rad – much less than that for sustained flutter oscillation of the bridge alone. Therefore, the STMD control system with variable damping is a promising way to control the flutter conditions of long-span suspension bridges.

12.8 Mechanical Measures for Buffeting Control

When the aerodynamic count measures are not able to reduce wind-induced buffeting response sufficiently, mechanical measures may be considered. These can be passive, active, semi-active or hybrid control systems, as discussed for flutter control, and these control systems can also be applied to bridge decks, cables and towers. The use of tuned mass dampers for mitigating wind-induced buffeting responses of long-span cable-supported bridges has been discussed in detail in the references [56,57]. For a wide coverage of the concerned topic, this section introduces multiple pressurized tuned liquid column dampers [58] and semi-active tuned liquid column dampers [59] for mitigating buffeting responses in either the construction or completion stages.

12.8.1 Multiple Pressurized Tuned Liquid Column Dampers

The pressurized tuned liquid column damper (PTLCD) originates from tuned liquid column dampers (TLCD), as discussed in Section 12.7.1. It is a U-shaped container of uniform cross-sectional area, with liquid filled into the container and two chambers at its two ends filled with compressed air of static pressure P_0 , as shown in Figure 12.25.

When the damper experiences a vibration due to structural motion, the volume of the two end chambers varies due to liquid motion inside the PTLCD. The change in the volume of an air chamber leads to

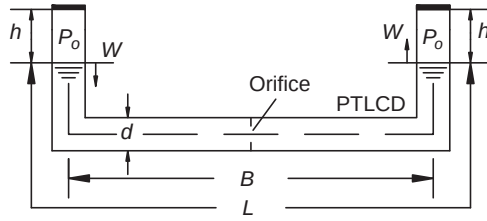


Figure 12.25 Schematic diagram of pressurized tuned liquid column damper (from [58]) (Reproduced with permission from Elsevier).

a variation of air chamber pressure and, thus, a pressure difference between the two chambers. The restoring force F_p , due to the pressure difference between the two chambers, can be expressed in the following form:

$$F_p = P \times A \quad (12.2)$$

where:

P is the net pressure between the two air chambers;

A is the cross-sectional area of container.

With an assumption that the variation of pressure and volume inside the air chamber is an isothermal process, the pressure and volume change can be described by Boyle's law, i.e.:

$$P_0 V_0 = P_R (V_0 - AW) = P_L (V_0 + AW) \quad (12.3)$$

where:

P_L and P_R are the pressure in the left and right air chamber of PTLCD respectively;

V_0 is the volume of air inside the chamber at the static position of liquid;

W is the liquid displacement of the PTLCD.

The restoring force F_p acting on the water at time t is then determined by:

$$PA = (P_R - P_L)A = P_o A \left[\left(1 - \frac{W}{h}\right)^{-1} - \left(1 + \frac{W}{h}\right)^{-1} \right] \quad (12.4)$$

where h is the air chamber height.

Equation 12.4 shows that the restoring force due to the pressure acting on the liquid inside the PTLCD is a non-linear function of water displacement. Iterations are generally required to solve Equation 12.4 in the time domain. However, if the ratio W/h is small, the restoring force in Equation 12.4 can be approximately expressed as a linear function of water displacement, i.e.:

$$PA \approx P_o A \left[\left(1 + \frac{W}{h}\right) - \left(1 - \frac{W}{h}\right) \right] = \frac{2P_o AW}{h} \quad (12.5)$$

Equation 12.5 shows that the restoring force due to the pressure inside the air chamber of the PTLCD is dependent on liquid displacement. For the case of a liquid column damper without air pressure, it becomes a traditional TLCD and the natural frequency of liquid in the TLCD is given by[51]:

$$\omega^2 = \frac{2 \rho_w A g}{\rho_w A L} = \frac{2g}{L} \quad (12.6)$$

where ρ_w is the density of liquid inside TLCD.

Note that the natural frequency of liquid motion in the TLCD depends solely on the liquid column length L of the TLCD. By applying a static pressure inside the two sealed air chambers at the two ends of TLCD, the natural frequency of liquid motion in the PTLCD becomes:

$$\omega^2 \approx \frac{2 \rho_w A g + \frac{2P_0 A}{h}}{\rho_w A L} = \frac{2g}{L} \left(1 + \frac{P_0}{\rho_w g h} \right) \quad (12.7)$$

Equation 12.7 shows that the natural frequency of liquid motion inside the PTLCD is determined not only by the length of liquid column but also the static pressure P_0 . For a given liquid column length, the natural frequency of the liquid motion inside the PTLCD can be increased by the factor $(1 + P_0/\rho_w g h)$, comparing with the traditional TLCD, which greatly facilitates the frequency tuning requirement. By re-arranging Equation 12.7, the static pressure P_0 inside the PTLCD is given by:

$$P_0 = \frac{\rho_w L h}{2} \left[\omega^2 - \frac{2g}{L} \right] \quad (12.8)$$

Equation 12.8 provides a way of determining the required static pressure P_0 after the frequency of the PTLCD and its liquid column length L are selected.

To consider the interaction between the liquid dampers and a long-span cable-supported bridge under wind excitation, it is expedient to derive the finite element model of the PTLCD or MPLCD, which is then incorporated into the finite element model of the bridge to form a coupled bridge-MPTLCD system. Let us consider the MPLCD, which consists of n_T small PTLCD units. Each small PTLCD unit is installed below the torsional centre of the bridge deck and at the locations where vibration amplitudes of the bridge in the lateral and torsional directions are the largest. The MPLCDs are connected to the transverse beams of the bridge deck by roller supports and simply supports, as shown in Figure 12.26.

Two additional nodes, named node 1 and node 2, are generated at the positions where the MPLCD is connected to the bridge. These nodes reflect the motion of the MPLCD interacted with the motion of the bridge. From a view point of practical use, the distance between the vertical columns B is the same for all PTLCD units. The axial deformation of the transverse beam between the two supports is assumed to be negligible and, hence, the lateral displacement of the PTLCD units is taken as x and the torsional displacement of the PTLCD unit is then determined by:

$$\theta = \frac{y_2 - y_1}{B} \quad (12.9)$$

where y_1 and y_2 are the vertical displacements of node 1 and node 2 respectively, as shown in Figure 12.26.

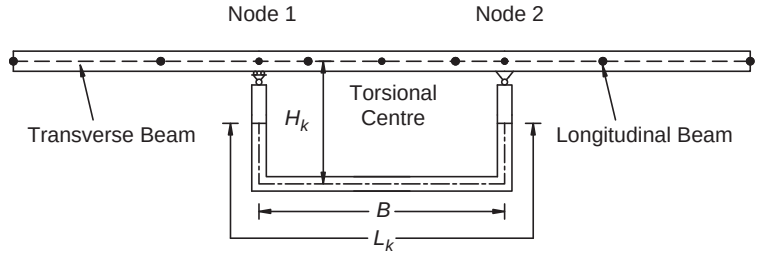


Figure 12.26 Connections between bridge deck and liquid damper (from [58]) (Reproduced with permission from Elsevier).

x shown in Figure 12.26 is the lateral displacement of node 2, which is not involved in the determination of the torsional displacement. The mass matrix $[M]$ and the stiffness matrix $[K]$ of the damper element can be obtained by formulating the Lagrangian of all PTLCDs used.

A real triple-tower cable-stayed bridge with an overall length of 1177 m, two main spans of 448 m and 475 m and two side spans of 127 m each, was used for a case study [58] to assess numerically the performance of MPTLCD for suppressing combined lateral and torsional vibration of the bridge (see Figure 12.27).

From the dynamic characteristics of this bridge, the dynamic lateral displacement response of the bridge deck was dominated by the first five lateral modes of vibrations, but the dynamic torsional displacement response was dominated by the first torsional mode of vibration only. Therefore, it was decided that for each main span, five MPTLCDs would be installed at the middle section (point A or point B) and would be tuned to the first five lateral frequencies of the bridge respectively, while one MPTLCD would be installed at the middle section (point A or point B) and would be tuned to the first torsional frequency of the bridge. The positions of point A and Point B are also shown in Figure 12.27. The standard deviation displacement and acceleration responses of the bridge deck are listed in Tables 12.1 and 12.2, respectively, for mean wind speeds of 20 m/sec and 50 m/sec at points A and B, with and without control.

It can be seen from Tables 12.1 and 12.2 that both the standard deviation displacement and acceleration responses in either lateral or torsional directions are reduced by the MPTLCDs effectively. The standard deviation displacement reduction in the lateral direction at point A reaches the level of 25% at a mean wind speed of 20 m/sec, and the level of 29% at a mean wind speed of 50 m/sec. The reduction of standard deviation torsional displacement response can reach the level of 21% at a mean wind speed

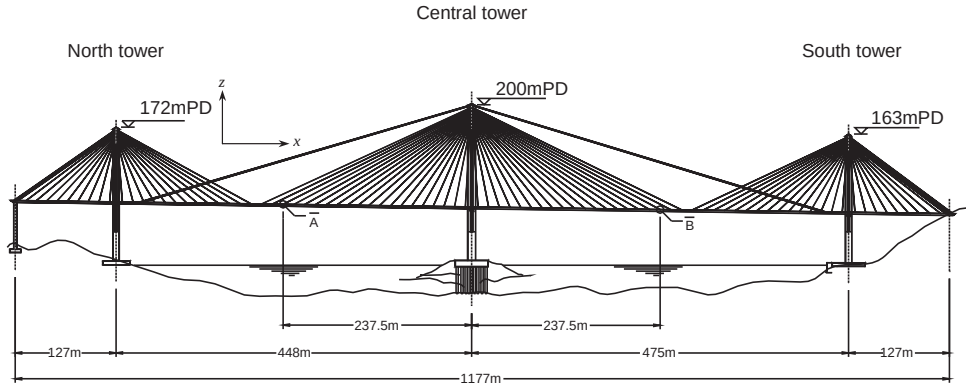


Figure 12.27 Configuration of long-span cable-stayed bridge (from [58]) (Reproduced with permission from Elsevier).

Table 12.1 Standard deviation of deck displacement with and without control (PTLCD) (from [58]) (Reproduced with permission from Elsevier)

Location	Lateral (m)		Torsional (rad)	
	A	B	A	B
Without control ($U_m = 20$ m/sec)	0.0028	0.0032	0.00 041	0.00 043
With control ($U_m = 20$ m/sec)	0.0021	0.0023	0.00 032	0.00 034
	(−25.0%)	(−28.1%)	(−22.0%)	(−20.9%)
Without control ($U_m = 50$ m/sec)	0.0287	0.0323	0.00 386	0.00 402
With control ($U_m = 50$ m/sec)	0.0204	0.0228	0.00 287	0.00 303
	(−28.9%)	(−29.4%)	(−25.7%)	(−24.6%)

Table 12.2 Standard deviation of deck acceleration with and without control (PTLCD) (from [58]) (Reproduced with permission from Elsevier)

Location	Lateral (m/sec^2)		Torsional (rad/sec^2)	
	A	B	A	B
Without control ($U_m = 20$ m/sec)	0.0137	0.0134	0.0048	0.0047
With control ($U_m = 20$ m/sec)	0.0107	0.0108	0.0037	0.0037
	(−21.9%)	(−19.4%)	(−22.9%)	(−21.3%)
Without control ($U_m = 50$ m/sec)	0.1375	0.1341	0.0467	0.0463
With control ($U_m = 50$ m/sec)	0.1058	0.1075	0.0357	0.0354
	(−23.1%)	(−19.8%)	(−23.6%)	(−23.5%)

of 20 m/sec and the level of 25% at a mean wind speed of 50 m/sec at point B. The standard deviations of lateral and torsional displacement responses of the bridge deck along the bridge axis at the mean wind speed of 50 m/sec are plotted in Figure 12.28, which shows that the maximum standard deviation of the displacement response of the bridge occurs near the midpoints of the two main spans and is reduced significantly by the MPTLCDs. However, the displacement responses of the deck near the towers are hardly reduced by the MPTLCDs. Further details on analytical models and numerical results can be found in the reference [58].

12.8.2 Semi-Active Tuned Liquid Column Dampers

PTLCD can be further developed to be a more robust PTLCD called semi-active tuned liquid column dampers (SATLCD) with frequency adaptability capacity. SATLCD is also a U-shaped container with uniform cross-sectional area (see Figure 12.29).

Liquid is filled into its container, and two chambers are filled with compressed air of static pressure P_0 . The control force is applied on the liquid in terms of a net external pressure between the two air chambers. The net pressure is regulated by the displacement and velocity of the liquid column in a prescribed way, so that the target natural frequency and damping of liquid motion inside the SATLCD can be easily achieved. The net pressure between the two air chambers, sensed by pressure transducers, is forced to follow or track the desired pressure determined by a computer in accordance with a given control algorithm and a targeted frequency. Any deviation from the desired pressure is fed back into the computer to take corrective action to adjust the servo valves.

Thus, the control system is continually monitoring and correcting pressure deviation to maintain the desired pressure acting on the liquid column and the targeted frequency. In practice, a number of SATLCDs are required for reducing vibration of a bridge. The control force is composed of two parts:

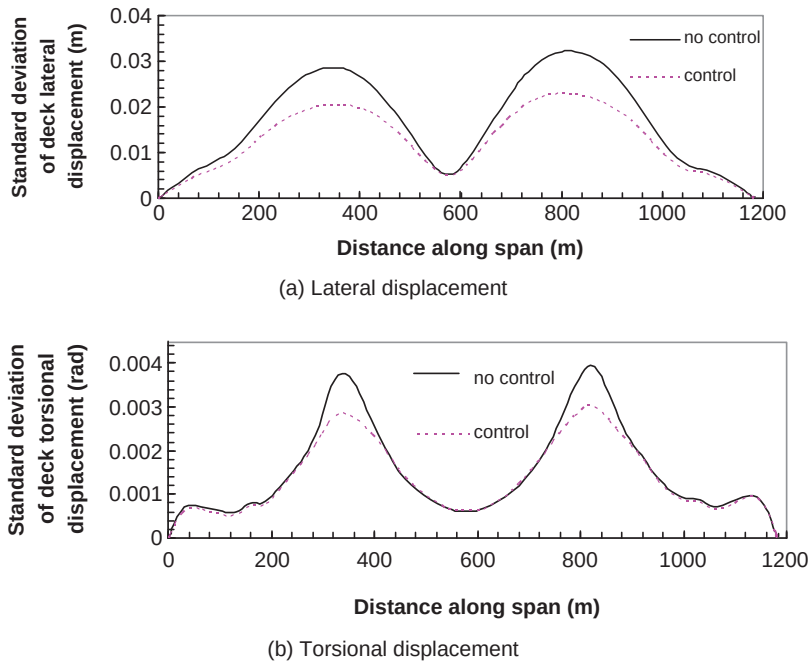


Figure 12.28 Comparisons of standard deviations of bridge deck displacements (from [58]) (Reproduced with permission from Elsevier).

one is displacement feedback force for the control of liquid oscillation frequency, while the other is velocity feedback force for the control of liquid damping. The displacement feedback control force $u_{1k}(t)$ in the k th SATLCD is given by:

$$u_{1k}(t) = S_k \times W_k(t) \tag{12.10}$$

where:

S_k is the constant displacement feedback gain of the k th SATLCD;
 $W_k(t)$ is the displacement of the liquid column.

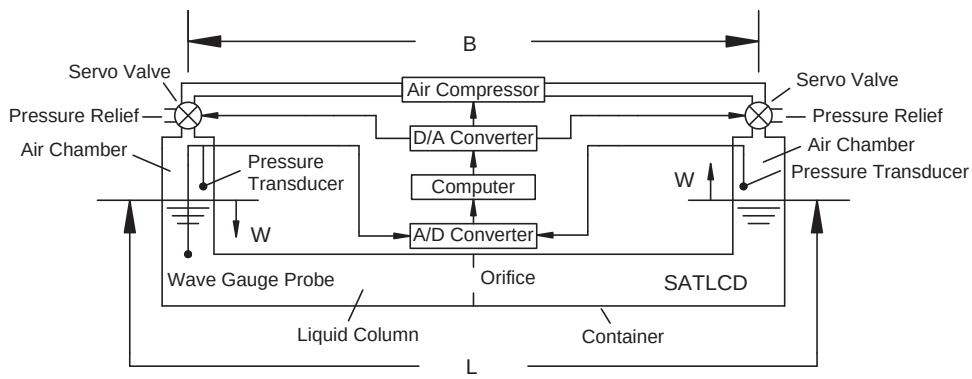


Figure 12.29 Schematic diagram of semi-active tuned liquid column damper (from [59]) (Reproduced with permission from Techno.Press).

The direction of the control force $u_{1k}(t)$ is in the same (opposite) direction as the liquid displacement $W_k(t)$ when the constant displacement feedback gain is positive (negative). With the control force expressed by Equation 12.10, the circular natural frequency, ω_k , of liquid motion in the k th SATLCD, can be determined by:

$$\omega_k^2 = \frac{2g}{L_k} + \frac{S_k}{\rho_w A_k L_k} \quad (12.11)$$

where:

L_k is the length of the k th SATLCD;

A_k is the cross-sectional area of the k th SATLCD;

ρ_w is the density of liquid inside the k th SATLCD.

For a targeted frequency of the liquid damper, it can easily be seen from Equation 12.11 that the liquid column length of the k th SATLCD is given by:

$$L_k = \frac{2g}{\omega_k^2} + \frac{S_k}{\rho_w A_k \omega_k^2} \quad (12.12)$$

Clearly, the liquid column length can be increased or decreased by adjusting the constant displacement feedback gain while keeping its frequency unchanged. The SATLCD is therefore more flexible than the traditional TLCD, in which S_k is equal to zero and there is no way for changing the liquid column length if the frequency is given. Once the frequency and length of the liquid column are decided, the required constant displacement feedback gain of the k th SATLCD can be determined by:

$$S_k = m_k \left[\omega_k^2 - \frac{2g}{L_k} \right] \quad (12.13)$$

where m_k is the mass of the k th SATLCD.

Malfunction of a SATLCD may result from excessive liquid motion when the bridge is subjected to high winds. An on-off control algorithm is therefore employed to make sure that liquid motion inside the SATLCD is within the tolerable limit. This velocity feedback control force is regulated by manipulating the pressure in accordance with the on-off control strategy as follows:

$$u_{2k}(t) = \begin{cases} \frac{1}{2} \rho_w A_k \delta_m |\dot{W}_k| \dot{W}_k & \text{if } (W_k > K_p \times W_a) \text{ and } (\dot{W}_k \times W_k > 0) \\ 0 & \text{otherwise} \end{cases} \quad (12.14)$$

where:

W_a is the tolerable liquid displacement of the k th SATLCD (see Equation 12.14);

K_p is an adjusting factor less than or equal to 1;

δ_m is the head loss coefficient for providing sufficient damping to the k th SATLCD and depends on $K_p W_a$.

Since the space inside the bridge deck is not large enough to provide a sufficiently long vertical column, the adoption of this particular control strategy (Equation 12.14) is to avoid the overflow of liquid inside the damper, particularly when the bridge under high wind speed. The non-linear damping force selected is to correlate this additional damping force with the passive damping force in term of head loss coefficient for easy manipulation. The control force is so selected that the additional damping is provided to the liquid when the liquid is continually increased beyond a certain level ($K_p W_a$). The factor K_p decides how the liquid displacement is close to the tolerable liquid displacement and when the additional damping control force should be added to the damper.

In other words, a larger value of the factor K_p would require a large value of δ_m to suppress the liquid motion as it gets closer to the tolerable liquid displacement than when there is a smaller value of K_p . The total control force acting on the k th SATLCD is then the sum of the control force, u_{1k} , based on the feedback of liquid displacement and the control force, u_{2k} , based on the feedback of liquid velocity:

$$u_k = \begin{cases} S_k W_k + \frac{1}{2} \rho_w A_k \delta_m |\dot{W}_k| \dot{W}_k & \text{if } (W_k > K_p \times W_a) \text{ and } (\dot{W}_k \times W_k > 0) \\ S_k W_k & \text{otherwise} \end{cases} \quad (12.15)$$

The desired control force acting on the liquid column can be provided by regulating the air pressure in the right chamber with respect to the air pressure in the left chamber, to obtain a net pressure $P_k(t)$. The relation between the net pressure and the control force can be expressed as:

$$u_k = P_k(t) \times A_k \quad (12.16)$$

The net pressure $P_k(t)$ in the k th SATLCD can be obtained from Equation 12.15 as:

$$P_k(t) = \begin{cases} \frac{S_k W_k}{A_k} + \frac{1}{2} \rho_w \delta_m |\dot{W}_k| \dot{W}_k & \text{if } (W_k > K_p \times W_a) \text{ and } (\dot{W}_k \times W_k > 0) \\ \frac{S_k W_k}{A_k} & \text{otherwise} \end{cases} \quad (12.17)$$

Inside the k th SATLCD, the air pressure in the left chamber P_L and in the right chamber P_R is then determined, respectively, by:

$$P_L = P_0 - \frac{P_k(t)}{2} \quad (12.18a)$$

$$P_R = P_0 + \frac{P_k(t)}{2} \quad (12.18b)$$

To consider the interaction between semi-active liquid column dampers and a long-span cable-stayed bridge under wind excitation, it is expedient to derive the finite element model of the liquid dampers so that it can be incorporated into the finite element model of the bridge. This procedure is similar to that for MPTLCD. Let us consider a total of n_T units of SATLCD installed below the torsional centre of the bridge deck and at the locations where vibration amplitudes of the bridge in the lateral and torsional directions are the largest. The SATLCD units are connected to the transverse beams of the bridge deck by roller supports and simple supports in a similar way to the PTLCD, as shown in Figure 12.26. Two additional nodes, node 1 and node 2, are generated at the positions where the SATLCD units are connected to the bridge. These two additional nodes reflect the motion of the SATLCD units interacted with the motion of the bridge. From a view point of practical use, the distance between the two vertical columns B is the same for all SATLCD units. The axial deformation of the transverse beam between the two supports is assumed to be negligible and, hence, the lateral displacement of the SATLCD units is taken as x and the torsional displacement of the SATLCD units is then determined by Equation 12.9. The Lagrangian of all SATLCD units can be expressed as follows:

$$L_d = \sum_{k=1}^n \left[\frac{1}{2} m_k \dot{W}_k^2 + \frac{1}{2} m_k \dot{x}^2 + \frac{1}{2} I_k \dot{\theta}^2 + m_k \alpha_k \dot{W}_k \dot{x} + G_k \dot{W}_k \dot{\theta} + m_k \bar{H}_k \dot{x} \dot{\theta} \right. \\ \left. + m_k g \bar{H}_k \cos \theta - m_k g \alpha_k W_k \sin \theta - \frac{m_k g}{L_k} W_k^2 \cos \theta - \frac{S_k}{2} W_k^2 \right] \quad (12.19)$$

where:

I_k is the second moment of the liquid mass inside the k th SATLCD;

α_k is the liquid length ratio $= B_k/L_k$;

$\bar{H}_k$ is the distance from the mass center of liquid inside the k th SATLCD to the torsional (elastic) center of the bridge deck;

G_k is the first moment of the liquid mass in the k th SATLCD;

and the last term in Equation 12.19 is the potential energy of liquid due to the control force u_{1k} in the k th SATLCD.

In Equation 12.19, the effect of vertical acceleration of the bridge deck on the liquid motion is neglected because it is relatively small compared with the acceleration due to gravity. The expressions for I_k , $\bar{H}_k$ and G_k are given as follows:

$$I_k = m_k \left(\alpha_k \left[H_k^2 + \frac{B_k^2}{12} \right] + (1 - \alpha_k) \left[H_k^2 + \frac{B_k^2}{4} - \frac{H_k(L_k - B_k)}{2} + \frac{(L_k - B_k)^2}{12} \right] \right) \quad (12.20a)$$

$$\bar{H}_k = H_k - \frac{(L_k - B_k)^2}{4L_k} \quad (12.20b)$$

$$G_k = m_k \alpha_k \left(H_k + \frac{L_k - B_k}{2} \right) \quad (12.20c)$$

where H_k is the vertical distance between the centerline of the horizontal part of the container and the torsional center of the bridge deck.

Equation 12.19 is subjected to the condition that the liquid should be fully retained in the horizontal part of the SATLCD and, thus, the following equation should always be satisfied:

$$W_k \leq \frac{L_k - B_k}{2} - \frac{d_k}{2} = W_a \quad (12.21)$$

where d_k is the thickness of the liquid column in the k th SATLCD.

The entries of mass matrix m_{ij} and stiffness matrix k_{ij} of the SATLCD (damper element) can be determined by:

$$m_{ij} = \frac{\partial}{\partial \dot{q}_j} \left(\frac{\partial L_d}{\partial \dot{q}_i} \right) \quad (12.22a)$$

$$k_{ij} = - \frac{\partial}{\partial q_j} \left(\frac{\partial L_d}{\partial q_i} \right) \quad (12.22b)$$

$$[q]^T = [x \quad y_1 \quad y_2 \quad W_1 \quad \dots \quad W_N] \quad (12.22c)$$

After some manipulations, the mass matrix of the damper element $[M]$ and the stiffness matrix of the damper element $[K]$ can be written as:

$$[M] = \begin{bmatrix} M_{11} & M_{12} \\ M_{12}^T & M_{22} \end{bmatrix} \quad (12.23a)$$

$$[K] = \begin{bmatrix} K_{11} & K_{12} \\ K_{12}^T & K_{22} \end{bmatrix} \quad (12.23b)$$

where:

$$M_{11} = \frac{1}{B^2} \begin{bmatrix} m_d B^2 & -mB & -mB \\ -mB & I_d & -I_d \\ -mB & -I_d & I_d \end{bmatrix} M_{12} = \begin{bmatrix} m_1 \alpha_1 & m_2 \alpha_2 & \dots & m_N \alpha_N \\ -\frac{G_1}{B} & -\frac{G_2}{B} & \dots & -\frac{G_N}{B} \\ \frac{G_1}{B} & \frac{G_2}{B} & \dots & \frac{G_N}{B} \end{bmatrix} \quad (12.24a)$$

$$M_{22} = \text{diag}(m_1, m_2, \dots, m_N) \quad (12.24b)$$

$$K_{11} = \frac{mg}{B^2} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} K_{12} = \frac{g}{B} \begin{bmatrix} 0 & 0 & \dots & 0 \\ -m_1 \alpha_1 & -m_2 \alpha_2 & \dots & -m_N \alpha_N \\ m_1 \alpha_1 & m_2 \alpha_2 & \dots & m_N \alpha_N \end{bmatrix} \quad (12.24c)$$

$$K_{22} = \text{diag}(m_1 \omega_1^2, m_2 \omega_2^2, \dots, m_N \omega_N^2) \quad (12.24d)$$

where:

m_d is the total liquid mass of all SATLCD units;

I_d is the total second moment of the liquid mass of all SATLCD units with respect to the torsional center of the bridge deck;

$$m = \sum_{k=1}^N m_k \bar{H}_k.$$

The vertical motion of liquid inside the liquid column is taken into consideration by including the restoring force of liquid due to gravitational force in the vertical direction. The vertical inertia effect of liquid mass in the global finite element model is also considered by modeling it as lumped masses at the corresponding node.

In consideration of the features of a long-span cable-stayed bridge during construction, SATLCDs with frequency adaptability were investigated to suppress combined lateral and torsional vibration of a long-span cable-stayed bridge under different stages of cantilever construction [59]. A real long-span cable-stayed bridge, as shown in Figure 12.27, was selected as a case study. Five different construction stages of the concerned bridge were considered (see Figure 12.30).

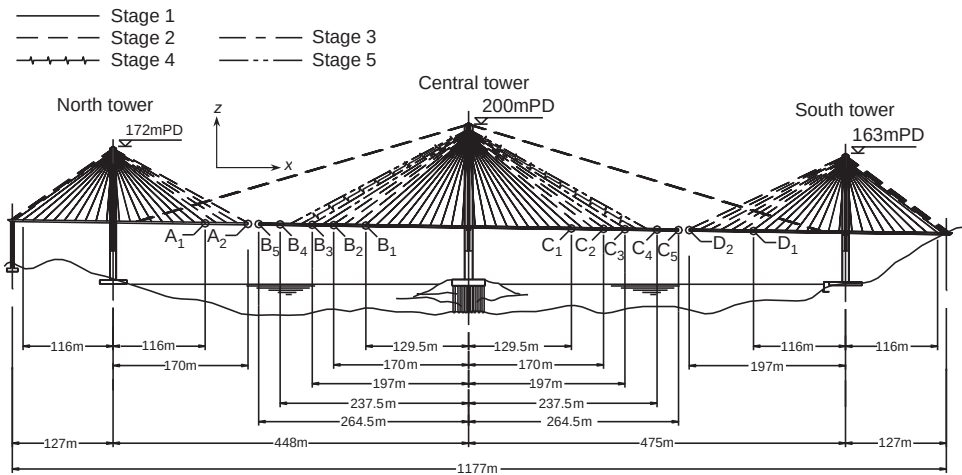


Figure 12.30 Configuration of long-span cable-stayed bridge under five different construction stages (from [59]) (Reproduced with permission from Techno.Press).

Figure 12.30 shows that the bridge under construction stage 1 is divided into three parts, which are erected simultaneously. Points A and D represent the locations at the tips of the cantilevers of the two side bridge decks. Points B and C represent the locations at the tips of the double cantilever of the bridge deck at the central tower part. Each part of the bridge deck is free at its two ends, and its transverse restraint is provided by the tower only. The deck-to-tower connections offer longitudinal and lateral displacement restraints, with essentially completely free rotation about all three axes, together with free vertical displacement. The span length of all incomplete bridge decks becomes longer at stage 2 and the central pylon is stabilized by the longitudinal stabilizing cables. The incomplete bridge deck, at two side spans, is fixed transversely and vertically at one end. The span of the incomplete bridge deck at the central tower is increased gradually from stage 3 to stage 5, at which point the bridge is almost completed.

The performance of SATLCD was examined by studying the double deck displacement and acceleration at points B and C. The results for the mean wind speeds at 20 m/sec and 50 m/sec are tabulated in Tables 12.3 and 12.4.

It can be seen from Tables 12.3 and 12.4 that both the standard deviation displacement and acceleration responses in either lateral or torsional direction are reduced by the SATLCD effectively. The standard deviation displacement reduction in the lateral direction reaches the level of 43% at a mean wind speed of 20 m/sec and the level of 29% at a mean wind speed of 50 m/sec. The reduction in standard deviation torsional displacement can reach 15% at a mean wind speed of 20 m/sec and 18% at a mean wind speed of 50 m/sec. The standard deviations of lateral, vertical and torsional displacement responses of the bridge deck along the bridge longitudinal axis at a mean wind speed of 20 m/sec are plotted in Figure 12.31.

It can be seen that the maximum standard deviation of the displacement response of the bridge deck occurs at the tips of the two cantilevers. The lateral displacement response of the bridge deck in the

Table 12.3 Standard deviation of deck displacement with and without control (SATLCD) (from [59]) (Reproduced with permission from Techno.Press)

Location	Lateral (m)		Vertical (m)		Torsional (rad)	
	B	C	B	C	B	C
Without control ($U_m = 20$ m/sec)	0.29 746	0.29 755	0.37 318	0.37 437	0.004 041	0.003 652
With control ($U_m = 20$ m/sec)	0.16 782	0.16 739	0.37 243	0.37 459	0.003 176	0.003 094
	(-43.6%)	(-43.7%)	(-0.2%)	(0.06%)	(-21.4%)	(-15.3%)
Without control ($U_m = 50$ m/sec)	1.87 074	1.87 170	2.45 735	2.46 240	0.031 593	0.029 666
With control ($U_m = 50$ m/sec)	1.32 582	1.32 442	2.44 805	2.45 722	0.023 934	0.024 204
	(-29.1%)	(-29.2%)	(-0.38%)	(-0.21%)	(-24.2%)	(-18.4%)

Table 12.4 Control standard deviation of deck acceleration with and without control (SATLCD) (from [59]) (Reproduced with permission from Techno.Press)

Location	Lateral (m/sec ²)		Vertical (m/sec ²)		Torsional (rad/sec ²)	
	B	C	B	C	B	C
Without control ($U_m = 20$ m/sec)	0.05 392	0.05 410	0.42 710	0.41 693	0.012 370	0.011 126
With control ($U_m = 20$ m/sec)	0.04 027	0.04 049	0.41 139	0.40 715	0.009 073	0.008 924
	(-25.3%)	(-25.2%)	(-3.68%)	(-2.35%)	(-26.6%)	(-19.8%)
Without control ($U_m = 50$ m/sec)	0.42 718	0.42 861	3.12 393	3.05 544	0.101 623	0.095 820
With control ($U_m = 50$ m/sec)	0.34 159	0.34 373	3.00 655	2.96 235	0.077 574	0.078 167
	(-20.0%)	(-19.8%)	(-3.76%)	(-3.05%)	(-23.7%)	(-18.4%)

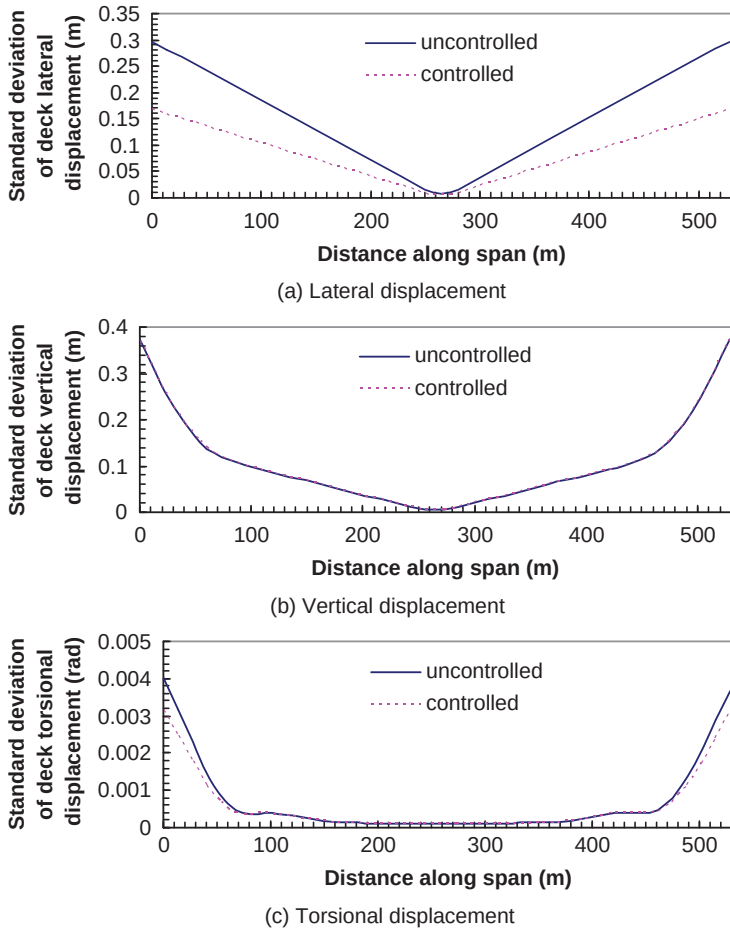


Figure 12.31 Standard deviations of deck displacements of the bridge under construction stage 5 (from [59]) (Reproduced with permission from Techno.Press).

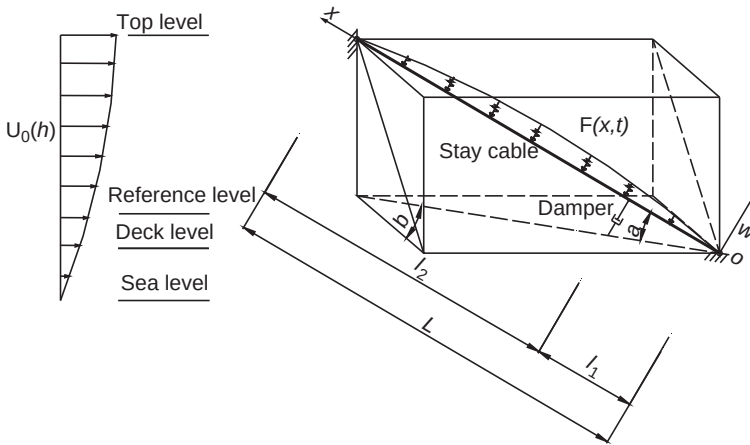
uncontrolled case increases linearly with increasing distance from the central tower, but the torsional displacement in the uncontrolled case is fairly small and increases suddenly near the tip of the cantilever. This may imply that the torsional stiffness of the deck near the tip of the cantilever is smaller. The lateral displacement response of the whole bridge deck is reduced effectively by the SATLCD. However, for reduction of torsional displacement, only the part with significant torsional vibration can be reduced.

12.9 Mechanical Measures for Rain-Wind-Induced Cable Vibration Control

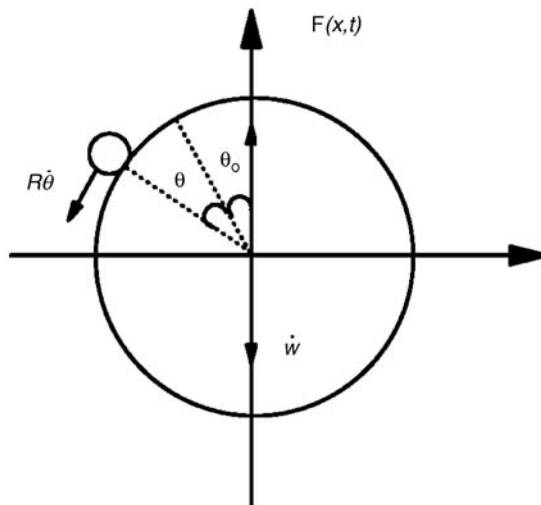
Similar to flutter and buffeting control, mechanical measures for rain-wind-induced cable vibration control can also be classified as passive control systems, active control systems, semi-active control systems and hybrid control systems, among which passive control systems (particularly viscous (oil) dampers) are widely used in practice throughout the world. In this connection, this session only introduces viscous dampers for mitigating rain-wind-induced cable vibration. The knowledge acquired from Chapter 5 will be useful in understanding the materials presented in this section.

The analytical model for predicting rain-wind-induced response of a stay cable with a viscous damper installed near the cable anchorage is presented based on the work of Zhou and Xu [60]. Most stay cables in a cable-stayed bridge are of small sag, in the order of 1% sag-to-length ratio, but with a high tension-to-weight ratio [61]. As a preliminary analytical study, the effects of sag and bending stiffness on cable vibration are neglected and only a taut inclined cable is considered, as shown in Figure 12.32a.

The inclination of the taut cable is denoted by angle α , and the yaw angle of incident wind is designated by angle β . To reduce possible cable vibration, discrete viscous dampers may be attached to the cable near its anchorage. The in-plane transverse motion $w(x,t)$ of a taut cable with



(a) the orientation of taut cable



(b) rivulet position

Figure 12.32 Modeling of rain-wind-induced vibration and control of a taut cable (from [60]) (Reproduced with permission from John Wiley & Sons Ltd.).

a viscous damper attached near its anchorage under rain-wind excitation can be described by the following partial differential equation of motion:

$$m \frac{\partial^2 w}{\partial t^2} - T \frac{\partial^2 w}{\partial x^2} + c \frac{\partial w}{\partial t} + F(x, t) + F_d(t) \delta(x - l_1) = 0, \quad 0 \leq x \leq L \quad (12.25)$$

with the boundary conditions $w(0, t) = w(L, t) = 0$,
where:

$w(x, t)$ is the in-plane transverse displacement normal to the cable axis at position x ;

m is the cable mass per unit length;

T is the cable tension force;

c is the internal cable damping coefficient;

$F(x, t)$ is the rain-wind-induced force on the cable;

$\delta(\bullet)$ is the Delta function;

l_1 is the length between the cable anchorage to the damper location;

L is the length of the cable;

$F_d(t)$ is the damper force on the cable.

When $F_d(t) = 0$, Equation 12.25 is reduced to the equation of motion of a taut cable under rain-wind excitation without control.

The fundamental assumption used in determining rain-wind-induced force on a stay cable is the quasi-steady approximation in consideration that rain-wind-induced cable oscillations are often very slow, as discussed in Chapter 5. By assuming that the upper rivulet vibrates circumferentially over the surface of the cable, rain-wind-induced force normal to the cable axis per unit length in the vertical direction, without considering turbulence and axis flow effects, can be expressed as:

$$F(x, t) = \frac{\rho D U_{rel}^2(x)}{2} [C_L(\phi) \cos(\phi + \theta + \theta_o) + C_D(\phi) \sin(\phi + \theta + \theta_o)] \quad (12.26)$$

where:

$U_{rel}(x)$ is the relative mean wind speed with respect to the cable motion;

D is the cable diameter;

ρ is the air density;

$\theta(x, t)$ is the dynamic angular displacement of upper rivulet with respect to the static position of upper rivulet $\theta_o(x)$ as shown in Figure 12.32b;

C_L is the lift coefficient of the cable with rivulet;

C_D is the drag coefficient of the cable with rivulet;

ϕ is the wind angle of attack.

Expansion of both drag and lift coefficients using the first three terms of Taylor's series leads to:

$$C_L(\phi) = A_0 + A_1 \phi + \frac{A_2}{2} \phi^2 + \frac{A_3}{6} \phi^3 \quad (12.27a)$$

$$C_D(\phi) = B_0 + B_1 \phi + \frac{B_2}{2} \phi^2 + \frac{B_3}{6} \phi^3 \quad (12.27b)$$

The substitution of Equations 12.27a and 12.27b, as well as the relationship between the relative mean wind speed $U_{rel}(x)$ and the incident mean wind speed $U_0(x)$ into Equation 12.26 and then the reservation of the linear terms of $\dot{w}$, θ , θ_o only yield:

$$F(x, t) = \frac{\rho D U_0(x)}{2} [R \dot{\theta}(x, t) \Gamma_1(x, t) + \dot{w}(x, t) \Gamma_2(x, t) + U_0(x) \theta(x, t) \Gamma_3(x, t)] \quad (12.28)$$

where:

R is the cable radius;

$\Gamma_1(\bullet)$, $\Gamma_2(\bullet)$ and $\Gamma_3(\bullet)$ are the aerodynamic functions which are not only the function of cable inclination angle, wind yaw angle, wind stagnation point and the mean wind speed via the static position of rivulet, but also the function of cable motion and rivulet motion, which may change the values of drag and lift coefficients.

Detailed information on the aerodynamic functions can be found in Section 5.4.2 in Chapter 5, in which ε is defined as the stagnation point influence factor. The stagnation point influence factor is introduced because, for rain-wind-induced vibration of a stay cable, wind direction is often not perpendicular to the cable axis, and the existence of an upper rivulet may change wind flow surrounding the cable. The stagnation point of incident wind on the surface of the cable is therefore not the same as that determined by the Bernoulli equation. No experimental results are available at present with respect to the position of stagnation point on a stay cable with upper rivulet. Nevertheless, numerical studies carried out by taking the stagnation point as a variable have indicated that the stagnation point influence factor could be taken as 0.4 [62].

In rain-wind-induced cable vibration, the motion of upper rivulet is actually coupled with the motion of cable. Therefore, in principle, the motion of upper rivulet should be taken as a variable and predicted together with the motion of cable. This, however, requires a very complicated analytical model for rain-wind-induced cable vibration. In this study, based on the observations from either field measurements or simulated rain-wind tunnel tests, the motion of upper rivulet, $\theta(x, t)$, is assumed to be harmonic and to follow the excited mode shape of cable as long as a steady-state cable vibration is concerned:

$$\theta(x, t) = a \cdot W(x) \sin(\omega_r t) \quad (12.29)$$

where:

a is the maximum amplitude of rivulet motion and is assumed to be a constant in this study;

$W(x)$ is the motion profile of rivulet along the cable axis and is assumed to be the same as the excited mode shape of cable;

ω_r is the frequency of rivulet motion and it is assumed to be the same as the natural frequency of cable corresponding to the excited mode shape.

For a taut cable without damper, the motion profile and frequency of rivulet corresponding to the i th mode of cable vibration are given by:

$$W(x) = \sin\left(\frac{\pi i}{L} x\right) \quad (12.30a)$$

$$\omega_i = i \omega_{o1} \quad (12.30b)$$

where $\omega_{o1} = \frac{\pi}{L} \sqrt{\frac{T}{m}}$ is the first circular natural frequency of the taut cable without damper.

For a taut cable with a viscous damper installed near its anchorage, its mode shapes are complex in general. Based on the asymptotic approximation [63], the motion profile of rivulet along the cable axis corresponding to the i th mode of vibration of the cable with a viscous damper near its anchorage can be approximately given by:

$$W(x) \cong \pm \sqrt{(1 - \Theta_{ei}) \sin^2\left(\frac{\pi i x}{L}\right) + H(x - l_1) \Theta_{ei} \sin^2(\pi i(x - l_1)/l_2)} \quad (12.31)$$

where:

$l_2 = L - l_1$ is the length from damper location to the top end of cable;

$H(\bullet)$ is the heaviside function.

$$H(x - l_1) = \begin{cases} 0 & 0 \leq x < l_1 \\ 1 & l_1 \leq x \leq L \end{cases} \quad (12.32)$$

Θ_{ci} is called the “clamping ratio” for the i th mode of vibration, which can be expressed as:

$$\Theta_{ci} \approx \frac{(\pi^2 \kappa)^2}{1 + (\pi^2 \kappa)^2} \quad (12.33)$$

where the non-dimensional damper damping parameter κ for the i th mode of vibration is defined as:

$$\kappa = \frac{c_L}{mL\omega_{o1}} i \frac{l_1}{L} \quad (12.34)$$

in which c_L is the damping coefficient of viscous damper.

The selection of positive or negative sign in Equation 12.31 depends on the phase angle ϕ_i :

$$\phi_i = \arctan \left(H(x - l_1) \sqrt{\Theta_{ci}} \sin \left(\frac{\pi i (x - l_1) / l_2}{\sqrt{(1 - \Theta_{ci})} \sin(\pi i x / L)} \right) \right) \quad (12.35)$$

Also based on the asymptotic approximation [63], the motion frequency of rivulet corresponding to the i th circular natural frequency of the cable with a viscous damper near its anchorage can be approximately given by:

$$\omega_r \cong i \left(1 + \Theta_{ci} \frac{l_1}{l_2} \right) \omega_{o1} \quad (12.36)$$

For a stay cable in a cable-stayed bridge, mean wind speed actually varies along with cable axis. Mean wind speed is zero at the sea level and increases with height above the sea level in the atmospheric boundary layer, as shown in Figure 12.32a. The mean wind speed profile could be approximately expressed by a power-law function:

$$U_0(h) = U_r \left(\frac{h}{h_r} \right)^{\alpha_1} \quad (12.37)$$

where:

$U_0(h)$ is the mean wind speed at height h above the sea level;

U_r is the mean wind speed at a chosen reference height h_r ;

α_1 is the exponent depending on the surface roughness.

Since the height h can be expressed as a function of x along the cable axis, Equation 12.37 can be rewritten as:

$$U_0(x) = U_0(h(x)) = U_r \left(\frac{h_0}{h_r} + \frac{x \sin \alpha}{h_r} \right)^{\alpha_1} \quad (12.38)$$

where h_0 is the height of lower anchorage of the cable above the sea level.

Linear viscous damper is considered in this study. Thus, the damper force acting perpendicularly on the cable at the position $x = l_1$ is given by:

$$F_d(t) = c_L \dot{w}(l_1, t) \quad (12.39)$$

For a taut cable with a linear viscous damper near its anchorage, Pacheco *et al.* [64] obtained the so-called universal design curve for the modal damping ratio in the cable through numerical complex-eigenvalue analysis. The mode shapes of the taut cable without damper were used as shape functions, and several hundred terms were required for adequate convergence in the solution, creating a computational burden. Main and Jones [65] recently used the clamped and unclamped mode shapes as shape functions in developing the numerical formulation of the eigenvalue problem of a taut cable with a linear or non-linear damper, which significantly reduces computational effort. This study extends their method to the case of rain-wind-induced vibration and control of a taut cable with a viscous damper near its anchorage.

The in-plane transverse displacement $w(x, t)$ of a taut cable with a linear viscous damper in Equation 12.25 can be assumed to be of the form:

$$w(x, \tau) = L[\Phi(x)^T \mathbf{z}(\tau)] = L[\Phi_o(x)^T \mathbf{z}_o(\tau) + \Phi_c(x)^T \mathbf{z}_c(\tau)] \quad (12.40)$$

where:

$\tau = \omega_{01} t$ is the non-dimensional time;

$\Phi(x) = [\Phi_o(x)^T \Phi_c(x)^T]^T$ is a vector of shape functions;

$\mathbf{z}(\tau) = [\mathbf{z}_o(\tau)^T \mathbf{z}_c(\tau)^T]^T$ is a vector of non-dimensional time-varying coefficients;

$\Phi_o(x)$ is a vector of unclamped mode shapes of cable, which is actually the vector of mode shapes of a taut cable without damper;

$\Phi_c(x)$ is a vector of clamped mode shapes of cable, which is the same as the vector of mode shapes of the taut cable when the damper becomes a clamp;

$\mathbf{z}_o(\tau)$ is the vector of the time-varying coefficient corresponding to the vector of unclamped mode shapes;

$\mathbf{z}_c(\tau)$ is the vector of the time-varying coefficient corresponding to the vector of clamped mode shapes.

The vector of the unclamped mode shapes of the taut cable is given by:

$$\Phi_o(x) = [\Phi_{o1}(x) \quad \Phi_{o2}(x) \quad \Phi_{o3}(x) \quad \dots]^T \quad (12.41a)$$

$$\Phi_{oi}(x) = \sin\left(\frac{\pi i x}{L}\right) \quad (12.41b)$$

The vector of the clamped mode shapes of the taut cable is expressed as:

$$\Phi_c(x) = [\Phi_{c1}(x) \quad \Phi_{c2}(x) \quad \Phi_{c3}(x) \quad \dots]^T \quad (12.42a)$$

$$\Phi_{ci}(x) = H(x - l_1) \sin\left[\frac{\pi i (x - l_1)}{L - l_1}\right] \quad (12.42b)$$

Substituting Equation 12.40 into Equation 12.25 and using Galerkin's method lead to the following equation of motion for rain-wind-induced vibration of a taut stay cable with a viscous damper:

$$\mathbf{M}\ddot{\mathbf{z}} + \mathbf{C}\dot{\mathbf{z}} + \mathbf{K}\mathbf{z} + \tilde{\mathbf{F}} = 0 \quad (12.43)$$

The mass matrix $\mathbf{M}$ and the stiffness matrix $\mathbf{K}$ in Equation 12.43 are given by:

$$\mathbf{M} = \begin{bmatrix} \Lambda_o & \Pi_{oc} \\ \Pi_{oc}^T & \Lambda_c \end{bmatrix} \quad (12.44a)$$

$$\mathbf{K} = \begin{bmatrix} \Lambda'_o & \Pi'_{oc} \\ \Pi'^T_{oc} & \Lambda'_c \end{bmatrix} \quad (12.44b)$$

The expression of each sub-matrix in Equations 12.44a and 12.44b can be found in the literature [63]. The damping matrix $\mathbf{C}$ in Equation 12.43 can be expressed as:

$$\mathbf{C} = 2\zeta_s \mathbf{M} + \mathbf{C}_0 + \mathbf{C}_a \quad (12.45)$$

in which ζ_s is the first structural modal damping ratio of the cable:

$$\mathbf{C}_0 = \frac{2c_L}{m(L\omega_{o1})} \begin{bmatrix} \Phi_o(l_1)\Phi_o(l_1)^T & 0 \\ 0 & 0 \end{bmatrix} \quad (12.46)$$

is the damping matrix due to the viscous damper; and

$$\mathbf{C}_a = \frac{\rho D}{L(m\omega_{o1})} \int_0^L \Phi(x) U_0(x) \Gamma_2(x, \tau) \Phi(x)^T dx \quad (12.47)$$

is the aeroelastic damping matrix due to the rain-wind-cable interaction. The normalized wind-rain-induced force in Equation 12.43 is given by:

$$\tilde{\mathbf{F}} = \frac{\rho D}{\pi^2 T} \int_0^L \Phi(x) U_0(x) [R\omega_{o1} \dot{\theta}(x, \tau) \Gamma_1(x, \tau) + U_0(x) \theta(x, \tau) \Gamma_3(x, \tau)] dx \quad (12.48)$$

The advantage of the numerical formulation presented above is that a good approximate solution could be obtained by using a small number of shape functions. As pointed out in [63], for a particular mode i of vibration of a taut cable with a damper near its anchorage, an accurate approximation could be obtained using just two shape functions: the i th unclamped mode shape $\Phi_{oi}(x)$ and the i th clamped mode shape $\Phi_{ci}(x)$. This feature greatly facilitates the numerical analysis of rain-wind-induced vibration and control of stay cable in a cable-stayed bridge.

Nevertheless, Equation 12.43 is a strongly non-linear equation, due to the physical interaction between cable, rivulet and wind. The rain-wind-induced force expressed in terms of Equation 12.48 is also an implicit function of cable motion and rivulet motion. The Runge-Kutta-Fehlberg method is therefore employed to find the numerical solutions [66]. This method is adaptive and easily implemented for the concerned numerical problem, in which a stay cable is divided into a number of sections along the cable axis and the numerical solutions are found at these sections for a series of time intervals, using the relevant unclamped and clamped shape functions. By using the Runge-Kutta-Fehlberg method, only six evaluations are required per step and local truncation error can be kept below a prescribed value by varying the step length.

Note that rain-wind-induced force depends on static rivulet position and cable motion, while the static rivulet position varies with wind speed. Rain-wind-induced force on the cable needs to be computed section-by-section, starting from the static rivulet position and cable motion at the previous time step.

Stay cable No.14 on the Meikonishi West Bridge, as investigated by Hikami and Shiraishi [42], is taken as an example in this study. The length of the cable is 75 m; the diameter of the cable is 0.14 m; the mass per unit length is 51 kg/m; the tension force in the cable is 1.147×10^6 N; the first structural damping ratio of the cable is estimated at 0.0011; the inclination of the cable is about 45° ; and the height of lower anchorage of the cable above the sea level h_0 is 45 m.

The field measurements carried out by Hikami and Shiraishi [42] recorded that when cable No. 14 was down-inclined along wind direction, with a wind yawed angle about 45° , the simultaneous occurrence of wind and rain caused excessive vibration of the cable in the first mode of vibration, with a natural frequency about 1 Hz. They also conducted a series of wind tunnel tests with rain conditions simulated, and reproduced rain-wind-induced cable vibration in their model tests. They observed from the model tests that the rivulet oscillated in a circumferential direction at the same period as the cable motion, and the amplitude of rivulet motion was around 10° . The drag and lift coefficient curves of the cylinder with artificial upper rivulet, and the rivulet static position curve as the function of mean wind speed, were also given by Yamaguchi [67] and Hikami and Shiraishi [42], respectively, and these are reproduced in Figures 12.33a and 12.33b, respectively.

In addition to the information provided above, the power exponent α_1 in the wind profile is selected as 0.16 for an open fetch. The stagnation point influence factor ε is selected as 0.4. The wind reference height h_r is 47 m above the sea level and the mean wind speed at the reference height is taken as a variable. The cable is then divided into 50 sections along the cable axis. The basic parameters used in the simulation of rain-wind-induced cable vibration are listed in Table 12.5. The analytical model and the numerical method proposed are finally applied to the cable with proper initial conditions, and an error tolerance of 10^{-4} , to compute rain-wind-induced displacement response of the cable mode by mode.

To investigate whether viscous dampers can mitigate rain-wind-induced cable vibration, a linear viscous damper is installed perpendicularly to the cable No.14 at the location of 2% of the cable length from the low cable anchorage. The non-dimensional damper damping parameter κ is taken as a variable, and the mean wind speed at the reference level is selected as 8.0, 9.0 and 10.0 m/sec, respectively. All other parameters of the cable and the rivulet remain the same as those mentioned above. Only the first mode of vibration of the cable is considered here.

Depicted in Figure 12.34 are rain-wind-induced maximum displacement responses of cable No. 14 against the non-dimensional damper damping parameter $\kappa \times \pi^2$ for the three mean wind speeds. It can be seen that for a given mean wind speed, there is an optimal value of $\kappa \times \pi^2$, by which the displacement response of the cable reaches its minimum. This optimal value is the same for three mean wind speeds and is around 1.0, i.e. the optimal non-dimensional damper damping parameter κ is equal to $1/\pi^2$. By using this optimal value, rain-wind-induced maximum displacement responses of the cable with the optimal damper are computed and compared with those of the cable without any damper for a series of mean wind speeds. The computed results are plotted in Figure 12.35, from which it can be seen that the damper with the optimal damping coefficient can effectively mitigate rain-wind-induced cable vibration. The global maximum displacement response of the cable is reduced from 0.22 m in the case without damper, to 0.03 m with the optimal damper.

Another parameter considered for evaluation of damper effectiveness is damper location. While all the other parameters of the cable and the rivulet are kept the same as those used above, the damper location is taken as a variable with the location at 2% and 4% of cable length from the low cable anchorage, respectively. The corresponding optimum non-dimensional damper damping parameter κ is $1/\pi^2$ for both locations. The computed rain-wind-induced maximum displacement responses of the cable are plotted in Figure 12.36 for a series of mean wind speeds at the reference level. Clearly, the further the damper is located away from the low anchorage, the more effective it is in reducing rain-wind-induced displacement response.

On the one hand, the above study shows that there is optimal damper damping coefficient by which rain-wind-induced displacement response of the cable can be reduced to its minimum, and that this optimal value is about $1/\pi^2$ in terms of the non-dimensional damper damping parameter κ . On the other

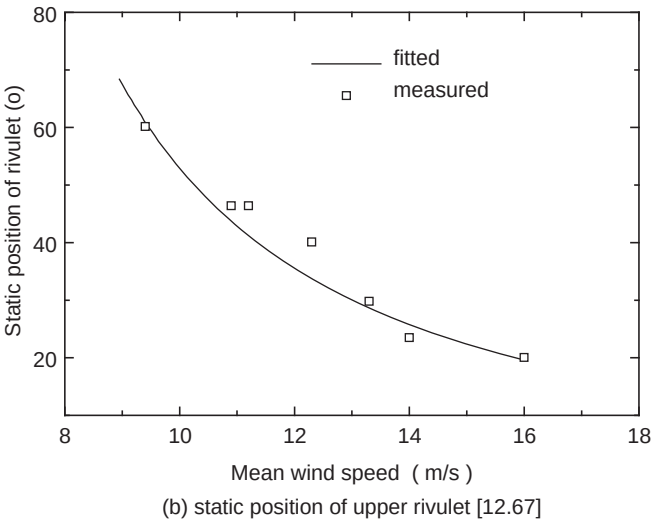
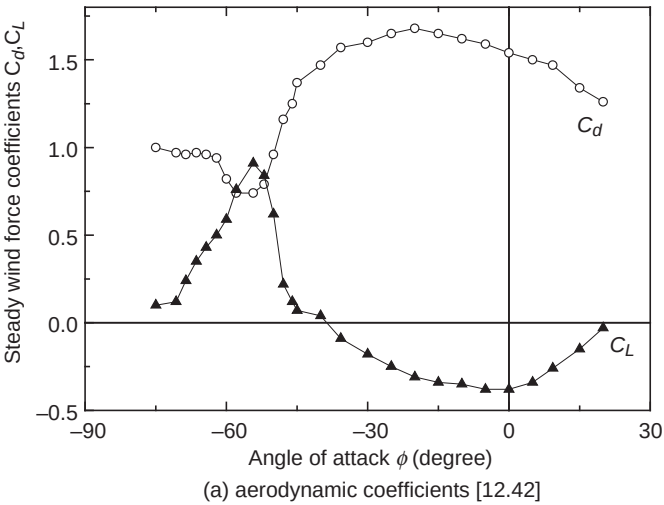


Figure 12.33 Aerodynamic properties of rain-wind-induced cylinder vibration (from [60]) (Reproduced with permission from John Wiley & Sons Ltd.).

hand, in current practice, the optimal damper damping parameter for a taut cable can be estimated using the so-called universal design curve obtained from a complex eigenvalue analysis [63,64,68], and the corresponding optimal modal damping ratio in the cable is then used to estimate the rain-wind-induced displacement response of the cable. It is thus interesting to know whether the optimal values and the rain-wind-induced displacement responses obtained from these two ways are the same.

Table 12.5 Basic parameters used in the simulation of rain-wind-induced cable vibration (from [60]) (Reproduced with permission from John Wiley & Sons Ltd)

Parameters	L (m)	D (m)	T ($\times 10^6$ N)	m (kg/m)	ζ_s (%)	α (°)	β (°)	h_0 (m)	h_r (m)	E	α_1
Values	75	0.14	1.147	51	0.1	45	45	45	47	0.4	0.16

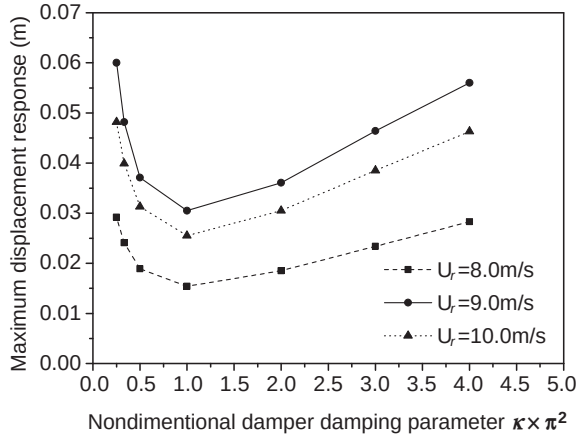


Figure 12.34 Rain-wind-induced maximum displacement responses of the cable with the non-dimensional damper damping parameter $\kappa \times \pi^2$ (from [60]) (Reproduced with permission from John Wiley & Sons Ltd.).

From the complex eigenvalue analysis of a taut cable with a linear viscous damper at the location l_1 from the low cable anchorage, the relationship between the i th modal damping ratio ζ_i in the cable and the non-dimensional damper damping parameter κ can be derived and expressed as:

$$\frac{\zeta_i}{(l_1/L)} = \frac{\pi^2 \kappa}{1 + (\pi^2 \kappa)^2} \quad (12.49)$$

When the non-dimensional damper damping parameter κ is taken as $1/\pi^2$, the modal damping ratio in the cable reaches its maximum value. Therefore, the optimal non-dimensional damper damping

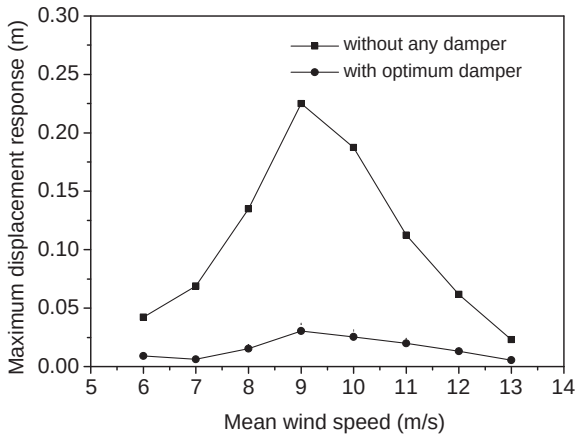


Figure 12.35 Comparison of rain-wind-induced maximum displacement responses between the cable with optimal damper and without any damper (from [60]) (Reproduced with permission from John Wiley & Sons Ltd.).

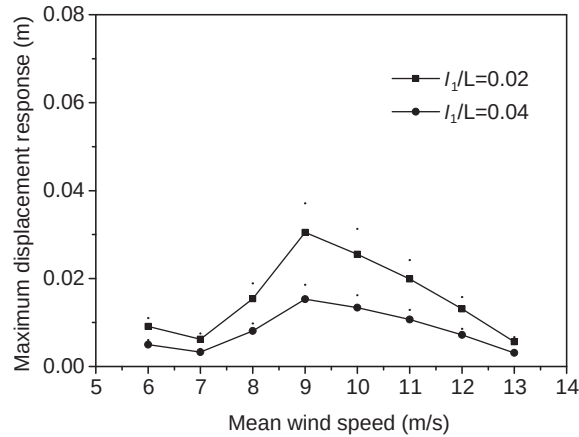


Figure 12.36 Effect of damper location on cable response (from [60]) (Reproduced with permission from John Wiley & Sons Ltd.).

parameters obtained from the two methods are actually the same. Furthermore, rain-wind-induced maximum displacement responses of cable No. 14 are computed for four cases:

- Case 1 is the cable with the optimal damper at the 2% location;
- Case 2 is the cable without damper, but with the additional optimal modal damping ratio obtained from the eigenvalue analysis of the cable with the optimal damper at the 2% location;
- Case 3 is the cable with the optimal damper at the 4% location;
- Case 4 is the cable without damper, but with the additional optimal modal damping ratio obtained from the eigenvalue analysis of the cable with the optimal damper at the 4% location.

The computed rain-wind-induced maximum displacement responses of cable No.14 are plotted in Figure 12.37 against different mean wind speeds at the reference level. It can be seen that the

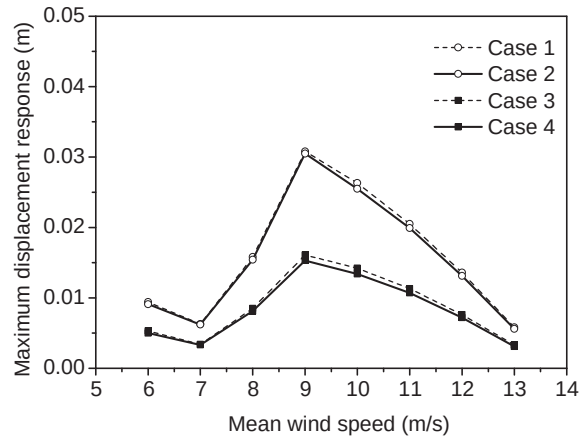


Figure 12.37 Comparison of rain-wind-induced maximum displacement responses of the cable for four cases (from [60]) (Reproduced with permission from John Wiley & Sons Ltd.).

rain-wind-induced maximum displacement responses of the cable computed from the two ways for a given damper location are almost the same, indicating that the current practice is acceptable.

12.10 Case Study: Damping Stay Cables in a Cable-Stayed Bridge

There are often hundreds of stay cables in a cable-stayed bridge, each of which features unique dynamic characteristics and requires a specific viscous damper to achieve the best control performance. If the viscous coefficient of the damper deviates from its optimum value, the modal damping ratio of the stay cable decreases rapidly. This requirement engenders many problems in the manufacture, implementation and maintenance of fluid dampers for a cable-stayed bridge. The use of MR with a semi-active control algorithm [69] may be an alternative, but some practical issues need to be solved before it can be accepted by the engineering professions. A new approach for damping vibration of stay cables in a cable-stayed bridge by using adjustable fluid dampers has been recently proposed by Xu and Zhou [70].

Different from a common passive fluid damper with a fixed number of orifices in its piston head, an adjustable fluid damper with shape memory alloy (SMA) actuators are installed inside the piston head in an adjustable fluid damper to control the number of orifices, in order to change damper parameters for the best control of a group of stay cables. After the optimum viscous coefficient of the damper of an adjustable fluid damper is achieved for a given cable, the damper then works as a passive fluid damper to maintain the practical merit of passive fluid dampers. A SMA actuator possesses at least two main characteristics:

1. It could control a small mechanical valve inside the damper piston to close or open an orifice.
2. the mechanical valve could be firmly locked at a prescribed position within a working temperature range.

A schematic diagram of a SMA actuator is shown in Figure 12.38a, and the SMA actuators installed in the piston of a prototype adjustable fluid damper are depicted in Figure 12.38b. There are two SMA wires, each of which is connected to one end of the valve block and fixed on the piston via two pulleys. By using an impulse current to heat one wire each time, the SMA wire will pull the valve block to open or close an orifice through the positioning plate and locating ball.

Two types of prototype adjustable fluid dampers were designed and manufactured: Type 1, with 1.5 mm diameter orifices for large viscous coefficient of the damper; and Type 2, with 1.8 mm diameter orifices for relatively small viscous coefficient of the damper. The two prototype dampers were extensively calibrated and tested, and the experimental results demonstrated that the damper performance could be well described by the following Maxwell model [71]:

$$F_d + \lambda_d \frac{dF_d}{dt} = c_d v \quad (12.50)$$

where:

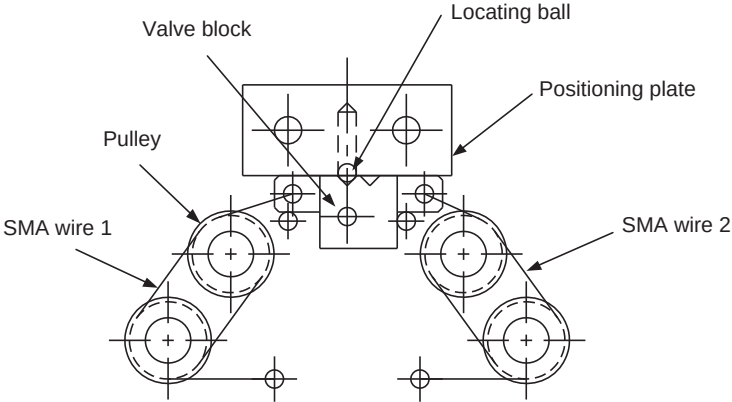
F_d is the damper force;

λ_d is the relaxation time constant;

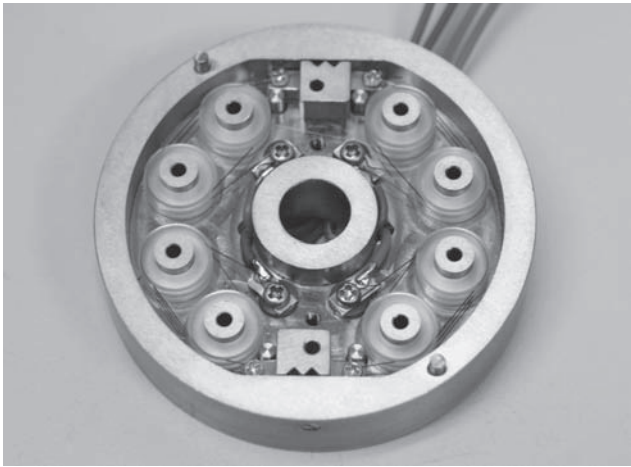
c_d is the viscous coefficient of the damper at zero frequency;

v is the velocity of the piston head.

Table 12.6 lists the experimental results of the viscous coefficient of the dampers and the relaxation time constants of the two types of adjustable fluid dampers developed. It can be seen that the relaxation time constant increases with increasing viscous coefficient of the damper at zero frequency. Both the viscous coefficient of the damper at zero frequency and the relaxation time constant become larger as



(a) Schematic diagram of SMA actuator



(b) Prototype of SMA actuator within a piston

Figure 12.38 Principle and prototype of SMA actuator (from [70]) (Reproduced with permission from Elsevier).

the opened orifice number n becomes smaller. More details on adjustable fluid dampers and the test results could be found in Li *et al.* [72].

In consideration that most stay cables in a cable-stayed bridge are of small sag (in the order of 1% sag-to-length ratio) but with a high tension-to-weight ratio, only a taut cable with an adjustable fluid damper installed near one of cable anchorage is considered, as shown in Figure 12.39.

The adjustable fluid damper described by the Maxwell model can be represented by a dashpot and a spring connected in series. The effect of the damper support is also considered in terms of a spring connected to the damper in series. If the length between the left cable anchorage and the damper is denoted as l_1 , then the length between the right cable anchorage and the damper is $l_2 = L - l_1$, in which L is the total length of the cable. In practice, the length l_1 is much smaller than the length l_2 .

The internal structural damping of a stay cable is very small compared with the damping provided by a properly-designed fluid damper, and it is thus neglected here. By considering the cable-damper

Table 12.6 The viscous coefficient of the dampers c_d and relaxation time constants λ_d of two types of adjustable fluid dampers against the opened orifice number n (from [70]) (Reproduced with permission from Elsevier)

n	Damper Type 1 (1.5 mm orifices)		Damper Type 2 (1.8 mm orifices)	
	c_d (Ns/m)	λ_d (sec.)	c_d (Ns/m)	λ_d (sec.)
10	137 000	0.0038	89 000	0
9	149 000	0.0053	101 000	0
8	173 000	0.0059	112 000	0.0013
7	198 000	0.0066	129 000	0.0021
6	228 000	0.0102	146 000	0.0056
5	267 000	0.0128	169 000	0.0069
4	332 000	0.0164	213 000	0.0098
3	446 000	0.0212	278 000	0.0158
2	621 000	0.0338	427 000	0.0228

system in two parts, x_1 and x_2 , using the damper position as a division, the free vibration of the cable-damper system in the transverse direction can be described by the following partial differential equation for each part of the cable:

$$m \frac{\partial^2 w_k(x_k, t)}{\partial t^2} = T \frac{\partial^2 w_k(x_k, t)}{\partial x_k^2} \quad (k = 1, 2) \quad (12.51)$$

where:

$w_k(x_k, t)$ is the transverse displacement of the cable at point x_k ;

x_k is the coordinate along the cable chord axis in the k th part;

the boundary conditions of the two parts of the cable are $w_1(0, t) = w_2(0, t) = 0$ for all t .

At the damper location, there is a discontinuity in cable slope, providing a transverse force matching the damper force F_d :

$$T \left[-\frac{\partial w_2}{\partial x_2} \Big|_{x_2=l_2} - \frac{\partial w_1}{\partial x_1} \Big|_{x_1=l_1} \right] = F_d \quad (12.52)$$

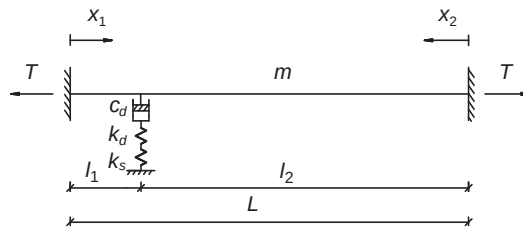


Figure 12.39 A taut cable with a fluid damper near its anchorage (from [70]) (Reproduced with permission from Elsevier).

Since the velocity of the damper piston is the same as the velocity of the cable at damper location, Equation 12.50 could be rewritten as:

$$F_d + \lambda \frac{dF_d}{dt} = c_d \frac{\partial w_1}{\partial t} \Big|_{x_1=l_1} \quad (12.53)$$

where:

$$\lambda = \lambda_d + \lambda_s = c_d/k_d + c_d/k_s \quad (12.54)$$

in which:

λ is the total relaxation time constant considering both damper stiffness and damper support stiffness;

λ_d is the relaxation time constant of damper itself;

λ_s is the equivalent relaxation time constant of damper support;

k_d is the damper stiffness;

k_s is the damper support stiffness.

Although the Maxwell damper was not considered previously [65,73], the frequency equation of the cable with an adjustable fluid damper can be obtained by simply substituting the complex mechanical impedance $c_d(1 + \lambda\omega_{o1}\chi)$ in place of the viscous coefficient c :

$$(1 + \lambda\omega_{o1}\chi)[\coth(\pi\chi l_1/L) + \coth(\pi\chi l_2/L)] + \frac{c_d}{\sqrt{Tm}} = 0 \quad (12.55)$$

where χ is the dimensionless eigenvalue that is complex in general.

For specific values of $c_d/\sqrt{Tm}$, l_1/L and $\lambda\omega_{o1}$, Equation 12.55 can be directly solved numerically to obtain a series of complex eigenvalues. In practice, the damper is often installed near the cable anchorage and l_1/L is rather small. Cable vibration mitigation also focuses on the first few modes of vibration only. Based on Equation 12.55 and the work of Main and Jones [65], the approximate (asymptotic) solution of the i th non-dimensional modal damping ratio of the cable-damper system can be found as:

$$\frac{\zeta_i}{l_1/L} \cong \frac{\pi^2 \kappa_i}{1 + (\pi^2 \kappa_i + i\lambda\omega_{o1})^2} \quad (12.56)$$

where:

$$\kappa_i \equiv \frac{c_d}{mL\omega_{o1}} i(l_1/L) \quad (12.57)$$

in which κ_i is termed the non-dimensional viscous coefficient of the damper for the i th mode of vibration.

It can also be found from Equation 12.56 that the maximum attainable damping ratio is $\zeta_{i,\max} \cong \frac{1}{2} \frac{l_1}{L} \left[\sqrt{(i\lambda\omega_{o1})^2 + 1} - i\lambda\omega_{o1} \right]$ when $\kappa_{i,opt} = \sqrt{1 + (i\lambda\omega_{o1})^2}/\pi^2$, where $\kappa_{i,opt}$ is the optimal non-dimensional viscous coefficient of the damper to achieve the maximum attainable modal damping ratio $\zeta_{i,\max}$.

Figure 12.40a displays the asymptotic solutions for $\frac{\zeta_i}{l_1/L}$ and κ_i for the first five modes of vibration, obtained by using Equation 12.56, for the cable-damper system with $l_1/L = 0.02$, $\lambda = 0.01$ (sec),

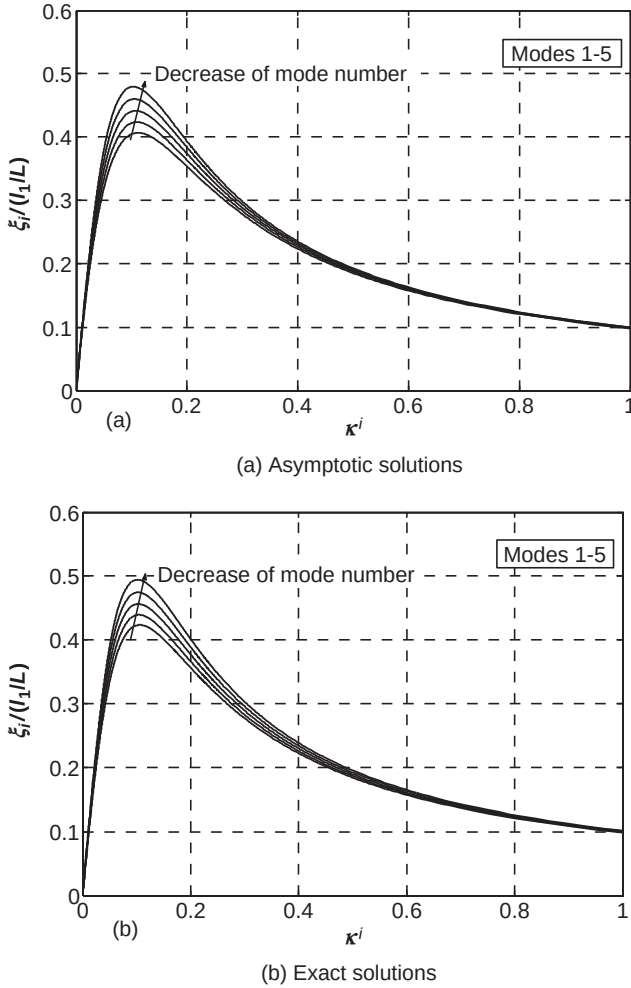
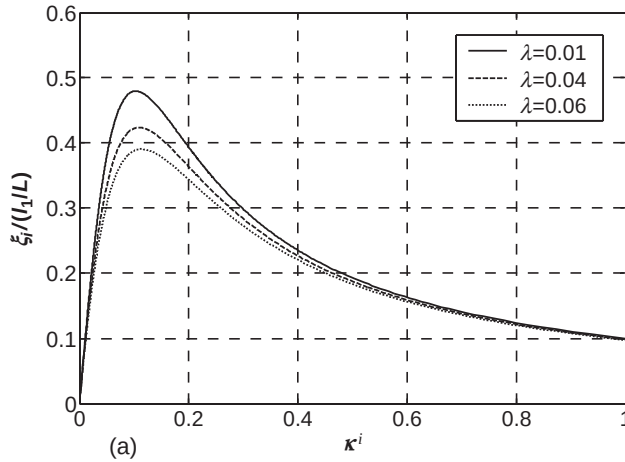


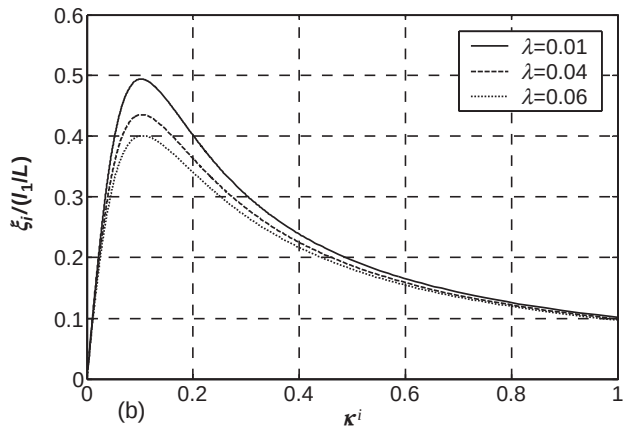
Figure 12.40 Cable modal damping ratio vs. non-dimensional viscous coefficient of the damper ($l_1/L=0.02$, $\lambda=0.01$ (sec), $\omega_{01}=4.152$ (rad/sec), $i=1-5$) (from [70]) (Reproduced with permission from Elsevier).

$\omega_{01}=4.152$ (rad/sec). Figure 12.40b depicts the exact solutions. It turns out that the attainable modal damping ratios obtained by Equation 12.56 are only slightly smaller than the exact solutions. However, both the asymptotic solutions and the exact solutions show that the attainable modal damping ratios are smaller for higher modes of vibration. This is due to the relaxation time constant, which is different from the case of the cable with a linear viscous damper, where the relaxation time constant is zero and all the curves overlap with each other.

The effects of relaxation time constant on attainable modal damping ratio can be found in Figure 12.41 for $\lambda=0.01, 0.04, 0.06$ (sec) and for $l_1/L=0.02$, $i=1$, $\omega_{01}=4.152$ (rad/sec), in which the results in Figure 12.41a are obtained from the asymptotic solution while those in Figure 12.41b are computed using the exact solution. The use of λ rather than the non-dimensional quantity $\lambda\omega_{01}$ is to be consistent with the experimental results provided in [72]. Again, the asymptotic results are very close to the accurate results



(a) Asymptotic solutions



(b) Exact solutions

Figure 12.41 Effects of relaxation time constant ($l_1/L = 0.02$, $\lambda = 0.01$ – 0.06 (sec), $\omega_{01} = 4.152$ (rad/sec), $i = 1$) (from [70]) (Reproduced with permission from Elsevier).

for different relaxation time constants λ . The maximum attainable non-dimensional modal damping ratio decreases as the relaxation time constant λ increases; such a reduction accounts for about 20% when the relaxation time constant $\lambda = 0.06$ (sec), comparing to $\lambda = 0$.

As the relaxation time constant is associated with damper stiffness and damper support stiffness, one may conclude that both the damper stiffness and the damper support stiffness will reduce the control effectiveness of the damper. Thus, it may be necessary to include the frequency dependence property in the analysis if a damper exhibits Maxwell characteristics and/or the damper support is not stiff enough.

From Equations 12.56 and 12.57, it can be seen that when a fluid damper is installed at a given location of a cable, the damper will have an optimum viscous coefficient of the damper that can achieve the maximum modal damping ratio in the cable. The optimum viscous coefficient of the damper,

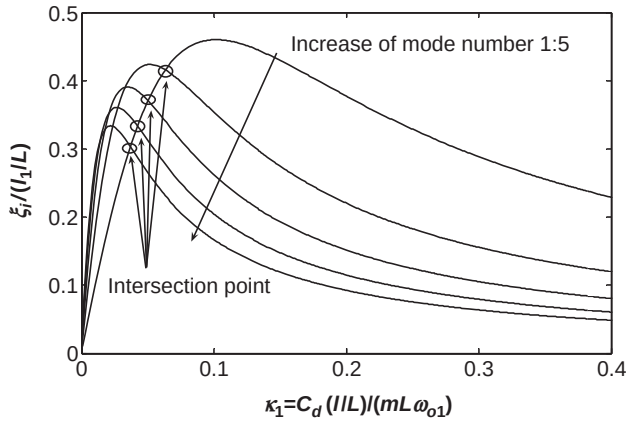


Figure 12.42 Intersection points for non-dimensional viscous coefficient of the damper ($l_1/L = 0.02$, $\lambda = 0.02$ (sec), $\omega_{01} = 4.152$ (rad/sec), $i = 1:5$) (from [70]) (Reproduced with permission from Elsevier).

however, depends on not only cable properties but also cable vibration mode. Let us consider a cable-damper system with $l_1/L = 0.02$, $\lambda = 0.02$ (sec), $\omega_{01} = 4.152$ (rad/sec).

Figure 12.42 plots the normalized modal damping ratios against the first non-dimensional viscous coefficient of the damper $\kappa_1 = c_d(l_1/L)/(mL\omega_{01})$ for the first five modes of vibration based on Equation 12.56. It can be seen that the lower mode of vibration has the higher maximum modal damping ratio. The corresponding optimum viscous coefficient of the damper c_d is also significantly different; the lower mode of vibration needs the larger optimum viscous coefficient of the damper.

Clearly, it is not possible for one fluid damper to achieve the maximum modal damping ratio for all modes of vibration of a given cable. Furthermore, as the optimum viscous coefficient of the damper depends on the properties and vibration modes of the cable and the location of the damper, it will be different for different cables. This implies that, ideally, every cable needs its own damper. However, this would cause many problems in the process of manufacture, installation and maintenance. Therefore, some compromise must be made and the grouping process is unavoidable in practice. This study proposes a most favorable design principle to select the favorable viscous coefficient of the damper for each cable and to group these favorable viscous coefficients of the damper according to the adjustable levels of one or two adjustable fluid dampers.

The most favorable design principle is to select the number of modes of vibration required to be damped for all stay cables in a cable-stayed bridge and at the same time, to comply with the requirement that the modal damping ratios in the concerned vibration modes for all stay cables should be greater than the least modal damping ratio limit ζ_{limit} . The determination of ζ_{limit} should consider many practical factors. For example, it may be selected as 0.5% damping ratio or 3% logarithmic decrement.

Adopting this favorable design principle, one should plot the $\zeta_i - \zeta_d$ curves similar to those in Figure 12.42 for each cable. Then, select the initial number of modes of vibration required to be damped, for instance i . Find the intersection point of the i th modal damping ratio curve and the first modal damping ratio curve for each cable from their $\zeta_i - \zeta_d$ curves. The viscous coefficient of the damper corresponding to the intersection point can then be regarded as the favorable viscous coefficient of the damper, and the corresponding modal damping ratio is regarded as the favorable modal damping ratio. If not all of the favorable modal damping ratios are greater than ζ_{limit} , the number of modes of vibration required to be damped should be reduced until all the favorable modal damping ratios are greater than ζ_{limit} . The favorable viscous coefficient of the damper for each cable can be found using Equation 12.56.

$$\text{Let } \frac{\zeta_1}{l_1/L} = \frac{\zeta_i}{l_1/L} \Rightarrow \frac{\pi^2 \frac{c_d}{mL\omega_{01}} (l_1/L)}{1 + \left[\pi^2 \frac{c_d}{mL\omega_{01}} (l_1/L) + \lambda\omega_{01} \right]^2} = \frac{\pi^2 \frac{c_d}{mL\omega_{01}} i(l_1/L)}{1 + \left[\pi^2 \frac{c_d}{mL\omega_{01}} i(l_1/L) + i\lambda\omega_{01} \right]^2} \quad (12.58)$$

then

$$c_{d,opt} = \frac{mL\omega_{01}}{\pi^2(l_1/L)} \left(\frac{1}{\sqrt{i}} - \lambda\omega_{01} \right) \quad (12.59)$$

where $c_{d,opt}$ is the favorable viscous coefficient of the damper.

The long-span cable-stayed bridge taken as a case study has a total length of 1596 m and a main span of 1018 m. The height of the two towers is nearly 300 m, measured from the base level to the top of the towers. The bridge tower is of a single composite column, and the stay cables are of the parallel wire strand type, made up of 7 mm wires. There are a total of 224 stay cables, and the length of the longest stay cable is about 540 m. For the sake of clear presentation, only one quarter of the stay cables of the bridge are considered, including 28 stay cables in the main span and 28 in the side span.

Figure 12.43 displays the computed first natural frequency, $f_1 = \omega_{01}/(2\pi)$, of each cable without any damper. The lowest first natural frequency is 0.244 Hz for cable No. 228 in the main span and the highest first natural frequency is 1.159 Hz for cable No. 102 in the side span. Figure 12.44 shows the ratio of the damper location to the original cable length, l_1/L , for each cable. The ratio ranges from 0.018 to 0.031. Let us assume that only one adjustable fluid damper is installed, perpendicular to the cable axis in the vertical plane near the low cable anchorage for each cable. As discussed before, the effect of the damper support stiffness may be considered. Figure 12.45 displays the damper support stiffness for each cable. It can be seen that the damper support stiffness increases as cable length becomes shorter.

The favorable design principle is used in this case study and the number of vibration modes is taken as two for all the stay cables concerned. The favorable design of adjustable fluid dampers is, in fact, an

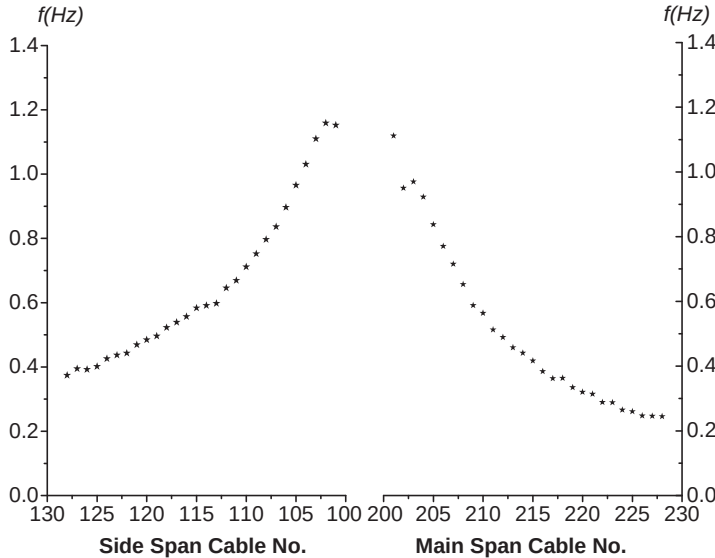


Figure 12.43 First natural frequencies of stay cables (from [70]) (Reproduced with permission from Elsevier).

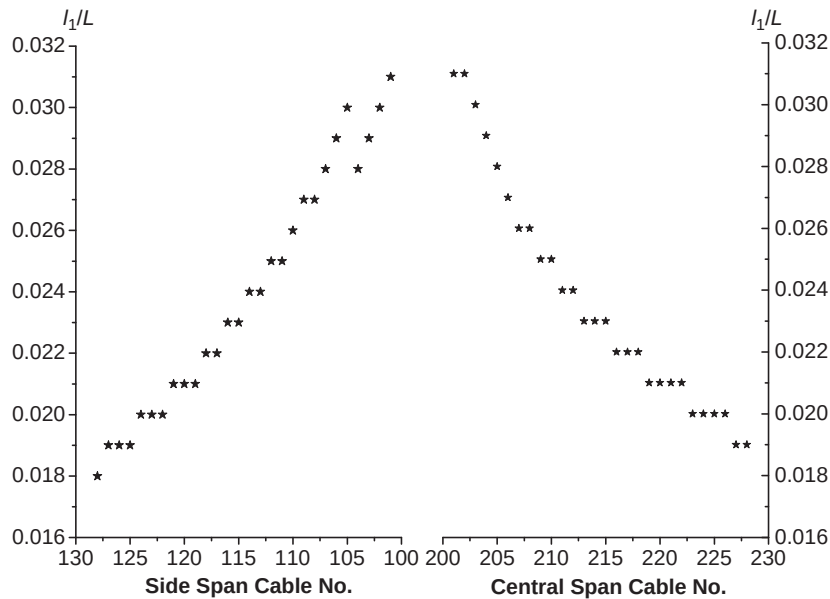


Figure 12.44 Ratios of damper location to cable length (from [70]) (Reproduced with permission from Elsevier).

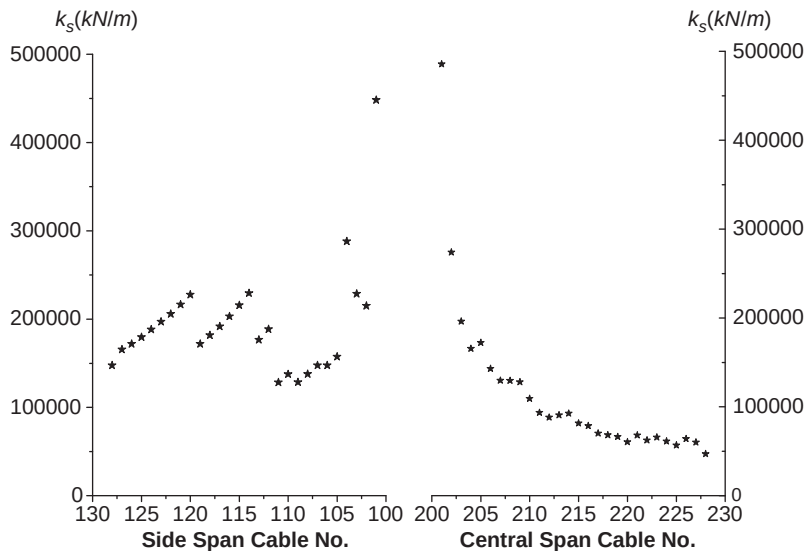


Figure 12.45 Damper support stiffness for each cable (from [70]) (Reproduced with permission from Elsevier).

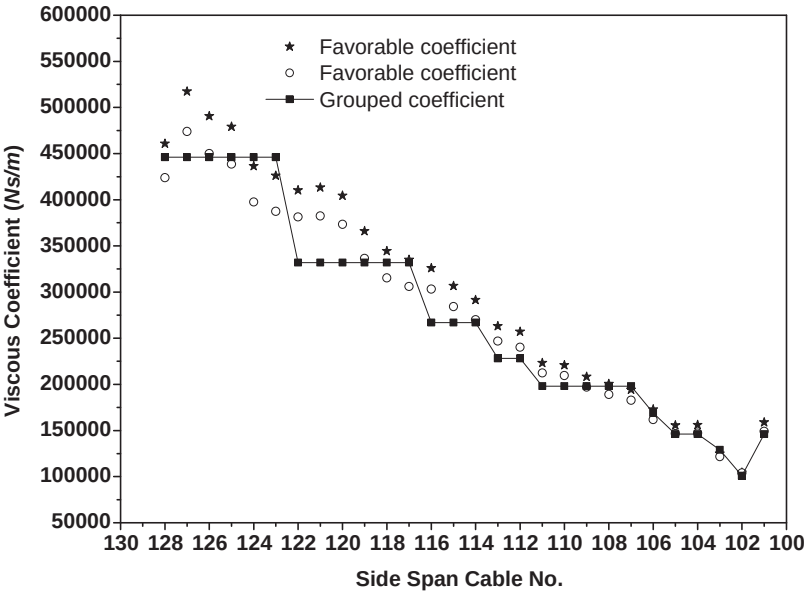


Figure 12.46 Viscous coefficient of the damper for stay cables in side span (from [70]) (Reproduced with permission from Elsevier).

iterative process. First, assume the damper is an ideally linear viscous damper and the damper support is perfectly rigid, so that the relaxation time constant is equal to zero. Then use Equation 12.59 to find the initial favorable viscous coefficient of the damper for each cable, as shown in Figure 12.46 for stay cables in the side span and in Figure 12.47 for stay cables in the main span.

Based on such information, adjustable fluid dampers are designed with enough adjustable levels to cover a wide range of viscous coefficient of the damper. The adjustable fluid dampers are manufactured and calibrated to find the viscous coefficient of the damper and relaxation time constants. In this case study, two types of prototype adjustable fluid dampers, which have ten orifices in the piston head – eight of them controllable, for achieving changes in damping at nine levels – were designed and manufactured: Type 1, with 1.5 mm diameter orifices for large viscous coefficient of the damper; and Type 2, with 1.8 mm diameter orifices for relatively small viscous coefficient of the damper. The two prototype dampers were extensively calibrated and tested.

The viscous coefficient of the damper and the relaxation time constants obtained from the tests are listed in Table 12.6. According to this, and the required initial favorable viscous coefficient of the damper as shown in Figures 12.46 and 12.47, it can be determined that the damper Type 2 should be used for those cables requiring a favorable viscous coefficient of the damper below 173 000 *Ns/m* and the damper Type 1 should be used for all other cables. One may also decide the adjustable levels of each damper, using the number of opened orifices in the damper to “best fit” the initial favorable viscous coefficient of the damper.

Figure 12.48 shows the grouping results of damper type and adjustable level. The next step is to calculate the total relaxation time constant for each cable, which includes both damper stiffness and damper support stiffness, as shown in Figure 12.49. The results indicate that, in this case study, the damper support stiffness is much larger than the damper stiffness.

Equation 12.59 is then used again to determine the favorable viscous coefficient of the damper, but with the relaxation time constants included. These results are plotted in Figures 12.46 and 12.47, and compared with the grouping results. If the comparison is not satisfactory, the grouping can be finely

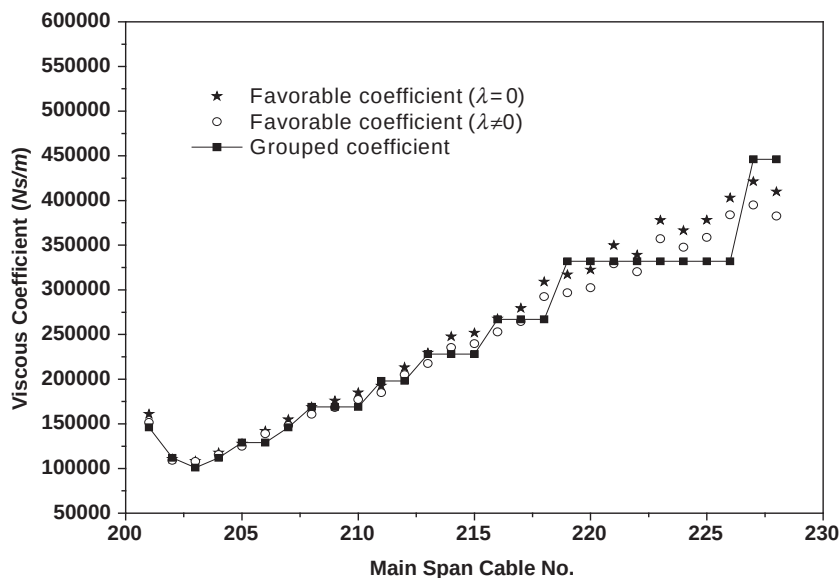


Figure 12.47 Viscous coefficient of the damper for stay cables in main span (from [70]) (Reproduced with permission from Elsevier).

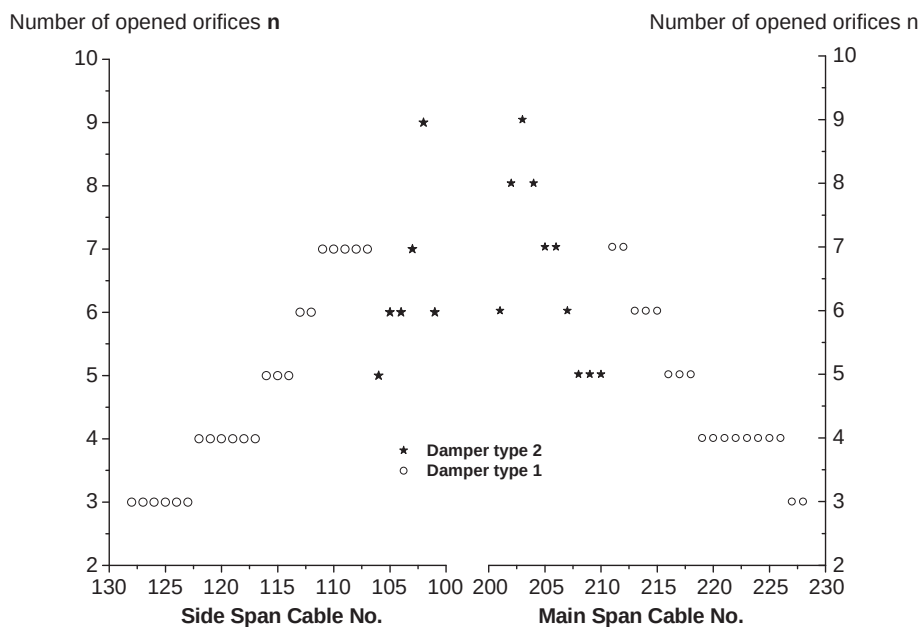


Figure 12.48 Grouping results of damper type and adjustable level (from [70]) (Reproduced with permission from Elsevier).

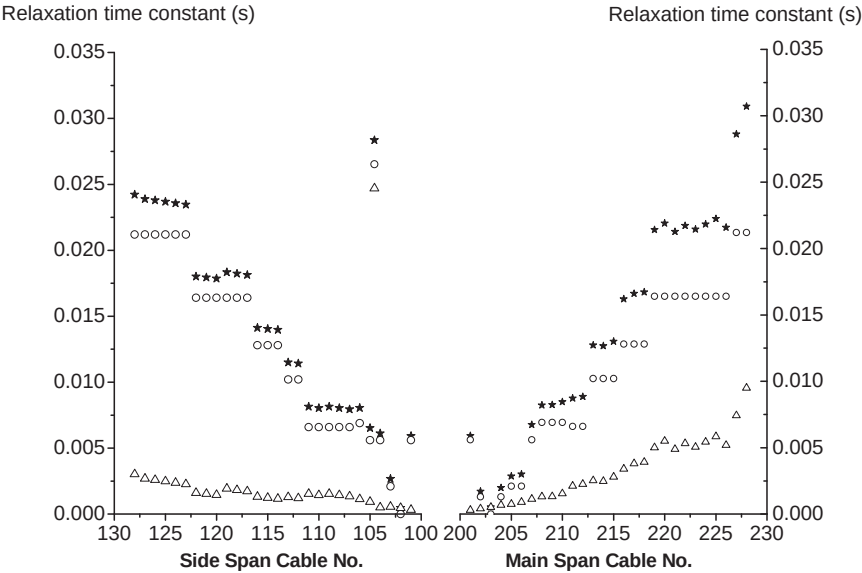


Figure 12.49 Relaxation time constants λ , λ_d , λ_s (from [70]) (Reproduced with permission from Elsevier).

tuned until both are close to each other. It can be seen from Figures 12.46 and 12.47 that some differences exist between the initial and final favorable viscous coefficient of the damper, but these are not very significant, because only the first two modes of vibration are considered in this case study.

Finally, based on the final grouping results, the favorable modal damping ratios in the first two modes of vibration for each cable can be recalculated and checked to see if they are greater than ζ_{limit} . The favorable modal damping logarithmic decrements obtained and shown in Figure 12.50 for this case study demonstrate that the modal damping logarithmic decrements in the first two modes of all the stay

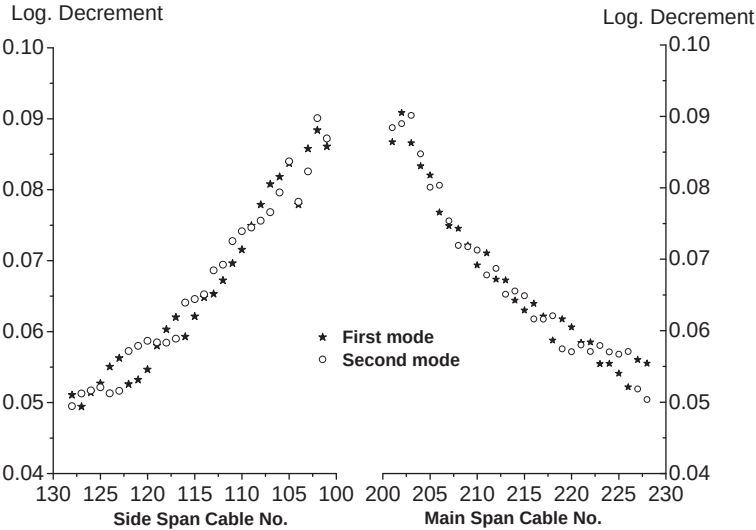


Figure 12.50 Modal damping logarithmic decrements in stay cables (from [70]) (Reproduced with permission from Elsevier).

cables concerned are greater than 4%. It should be pointed out, however, that the favorable modal damping logarithmic decrements shown in Figure 12.50 are obtained on the basis of the cable being assumed to be taut. For the first mode of vibration of the longest stay cable, sag effect may reduce the modal damping logarithmic decrement to some extent.

12.11 Notations

A	Cross-sectional area of container
A_k	Cross-sectional area of the k th SATLCD
a	Maximum amplitude of rivulet motion
B	Horizontal width of the water column of the TLCD
C	Damping matrix
C_a	Aeroelastic damping matrix due to the wind-rain-cable interaction
C_0	Damping matrix due to the viscous damper
C_D	Drag coefficient of the cable with rivulet
C_L	Lift coefficient of the cable with rivulet
c	Internal cable damping coefficient
c_d	Viscous coefficient of the damper at zero frequency
$c_{d,opt}$	Favorable viscous coefficient of the damper
c_L	Damping coefficient of viscous damper
D	Cable diameter
d_k	Thickness of the liquid column in the k th SATLCD
$F(x,t)$	Rain-wind-induced force on the cable
$F_d(t)$	Damper force on the cable
F_p	Restoring force
$\bar{F}$	Normalized rain-wind-induced force
f_1	First natural frequency
G_k	First moment of the liquid mass in the k th SATLCD
g	Acceleration due to gravity
H	Distance from the center line of the bottom tube of the TLCD to the pivot
$H(\bullet)$	Heaviside function
h	Air chamber height
h_r	Wind reference height
h_0	Lower anchorage height of the cable above the sea level
I_d	Total second moment of the liquid mass of all SATLCD units with respect to the torsional center of the bridge deck
I_k	Second moment of the liquid mass inside the k th SATLCD
I	Mass moments of inertia
K_p	Adjusting factor less than or equal to 1
$[K]$	Stiffness matrix
$\mathbf{K}$	Stiffness matrix
k_{ij}	Stiffness matrix of the SATLCD
k_d	Damper stiffness
k_s	Damper support stiffness
L	(i) Total length of water column (Equations (12.1a)–(12.1c)) (ii) Liquid column length (Equations 12.6 and 12.7) (iii) Cable length (Equation 12.25)
L_k	Length of the k th SATLCD

l_1	Length between the cable anchorage to the damper location
l_2	Length from damper location to the top end of cable
m	Cable mass per unit length (Equations 12.25 and 12.51)
m_d	Total liquid mass of all SATLCD units
m_k	Mass of the k th SATLCD
m_{ij}	Mass matrix of the SATLCD
$[M]$	Mass matrix
$\mathbf{M}$	Mass matrix
P	Net pressure between the two air chambers
P_L	Pressure in the left air chamber
P_R	Pressure in the right air chamber
P_0	Static pressure
R	Cable radius
S_k	Constant displacement feedback gain of the k th SATLCD
T	Cable tension force
t	Time
$U_{rel}(x)$	Relative mean wind speed with respect to the cable motion
$U_0(x)$	Incident mean wind speed
$U_0(h)$	Mean wind speed at height h above the sea level
U_r	Mean wind speed at a chosen reference height h_r
$u_{1k}(t)$	Displacement feedback control force in the k th SATLCD
u_{2k}	Control force based on the feedback of liquid velocity
V_0	Volume of air inside the chamber at the static position of liquid
W	(i) Relative motion of water to the container (Equations (12.1a–12.1c)) (ii) Liquid displacement (Equation 12.3)
W_a	Tolerable liquid displacement of the k th SATLCD
$W(x)$	Motion profile of rivulet along the cable axis
$w(x,t)$	In-plane transverse displacement normal to the cable axis at position x
x	Lateral displacement of node 2
y_1	Vertical displacement of node 1
y_2	Vertical displacement of node 2
$\mathbf{z}(\tau)$	Vector of non-dimensional time-varying coefficient
$\mathbf{z}_c(\tau)$	Vector of time-varying coefficient corresponding to the vector of clamped mode shapes
$\mathbf{z}_0(\tau)$	Vector of time-varying coefficient corresponding to the vector of unclamped mode shapes
$\bar{H}_k$	Distance from the mass center of liquid inside the k th SATLCD to the torsional (elastic) center of the bridge deck
ρ_w	Liquid density
ρ	Air density
α	Inclination angle of the taut cable
α_1	Exponent depending on surface roughness
α_k	Liquid length ratio
β	Yaw angle of incident wind
ε	Stagnation point influence factor
ω_k	Circular natural frequency
ω_r	Frequency of rivulet motion
ω_{01}	First circular natural frequency of the taut cable without damper
θ	Torsional displacement
λ	Total relaxation time constant considering both damper stiffness and damper support stiffness
λ_d	Relaxation time constant of damper itself
λ_s	Equivalent relaxation time constant of damper support

τ	Non-dimensional time
$\theta(x, t)$	Dynamic angular displacement of upper rivulet with respect to the static position of upper rivulet $\theta_0(x)$
$\zeta_{i, \max}$	Maximum attainable modal damping ratio
ζ_{limit}	Least modal damping ratio
χ	Dimensionless eigenvalue
v	Velocity of piston head
κ	Non-dimensional damper damping parameter
κ_i	Non-dimensional viscous coefficient of the damper for the i th mode of vibration
$\kappa_{i, \text{opt}}$	Optimal non-dimensional viscous coefficient of the damper
ϕ	(i) Wind angle of attack (Equations 12.26 and 12.27a, 12.27b) (ii) Phase angle (Equation 12.35)
δ_m	Head loss coefficient for providing sufficient damping to the k th SATLCD
$\delta(\bullet)$	Delta function
$\Phi(x)$	Vector of shape function
$\Phi_c(x)$	Vector of clamped mode shapes of cable
$\Phi_0(x)$	Vector of unclamped mode shapes of cable
Θ_{ci}	“Clamping ratio” for the i th mode of vibration
$\Gamma(\bullet)$	Aerodynamic function

References

1. Fujino, Y., Kimura, K., and Tanaka, H. (2012) *Wind Resistant Design of Bridges in Japan: Development and Practices*, Springer Verlag, Japan.
2. Fujino, Y. (2002) Vibration, control and monitoring of long-span bridges- recent research, developments and practice in Japan. *Journal of Constructional Steel Research*, **58** (1), 71–97.
3. Ostenfeld, K. and Larsen, A. (1992) Bridge engineering and aerodynamics, in *Aerodynamics of Large Bridges: Proceedings of the First International Symposium on Aerodynamics of Large Bridges* (ed. A. Larsen), Balkema, Rotterdam, The Netherlands, pp. 3–22.
4. Astiz, M.A. (1998) Flutter stability of very long suspension bridges. *Journal of Bridge Engineering*, ASCE, **3** (3), 132–139.
5. Branceleoni, F. (1992) The construction phase and its aerodynamic issues, in *Aerodynamics of Large Bridges: Proceedings of the First International Symposium on Aerodynamics of Large Bridges* (ed. A. Larsen), Balkema, Rotterdam, The Netherlands, pp. 147–158.
6. Wilde, K., Fujino, Y., and Prabis, V. (1996) Effects of eccentric mass on flutter of long span bridge. Proceedings of 2nd International Workshop on Structural Control, Hong Kong, pp. 564–574.
7. Xiang, H.F. (2005) *Modern Theory and Practice on Bridge Wind Resistance*, China Communications Press, Beijing (in Chinese).
8. Housner, G.W., Bergman, L.A., Caughey, T.K. et al. (1997) Structural control: Past, present and future. *Journal of Engineering Mechanics*, ASCE, **123** (9), 897–971.
9. Soong, T.T. and Dargush, G.F. (1997) *Passive Energy Dissipation Systems in Structural Engineering*, John Wiley & Sons, Inc., New York, N.Y.
10. Spencer, B.F. Jr. and Soong, T.T. (1999) New applications and development of active, semi-active and hybrid control techniques for seismic and non-seismic vibration in the USA. Proceedings of International Post-SMiRT Conference on Seismic Isolation, Passive Energy Dissipation and Active Control of Vibration of Structures, Cheju, Korea, August, pp. 23–25.
11. Soong, T.T. (1990) *Active Structural Control: Theory and Practice*, Longman Scientific, Essex, UK.
12. Spencer, B.F. Jr., Dyke, S.J., Sain, M.K., and Carlson, J.D. (1997) Phenomenological model of a magnetorheological damper. *Journal of Engineering Mechanics*, ASCE, **123** (3), 230–238.
13. Spencer, B.F. Jr. and Nagarajaiah, S. (2003) State of the art of structural control. *Journal of Structural Engineering*, ASCE, **129** (7), 845–856.
14. Xu, Y.L., Qu, W.L., and Ko, J.M. (2000) Seismic response control of frame structures using magnetorheological/electrorheological dampers. *Earthquake Engineering and Structural Dynamics*, **29** (5), 557–575.
15. Richardson, J.R. (1981) The Development of the Concept of the Twin Suspension Bridge, National Maritime Institute, NMIR125.
16. Sato, H., Toriumi, R., and Kusakabe, T. (1995) Aerodynamic characteristics of slotted box girders. Proceedings of Bridges into the 21st Century, Hong Kong.

17. Larsen, A. and Astiz, M.A. (1998) Aeroelastic consideration for the Gibraltar bridge feasibility study, in *Bridge Aerodynamics* (eds Larsen, A. and Esdahl, S.), Balkema, Rotterdam, pp. 165–173.
18. Matsumoto, M., Yoshizumi, F., Yabutani, T. *et al.* (1999) Flutter stabilization and heaving-branch flutter. *Journal of Wind Engineering and Industrial Aerodynamics*, **83** (1–3), 289–299.
19. Sato, H., Kusuhashi, S., Ogi, K., and Matsufuji, H. (2000) Aerodynamic characteristics of super long-span bridges with slotted box girder. *Journal of Wind Engineering and Industrial Aerodynamics*, **88** (2–3), 297–306.
20. Sato, H., Hirahara, N., Fumoto, K. *et al.* (2002) Full aeroelastic model test of a super long-span bridge with slotted box girder. *Journal of Wind Engineering and Industrial Aerodynamics*, **90** (12–15), 2023–2032.
21. Xiang, H.F. and Ge, Y.J. (2003) On aerodynamic limit to suspension bridges. Proceedings of the 11th International Conference on Wind Engineering, Texas, USA, pp. 65–80.
22. Simiu, E. and Miyata, T. (2006) *Design of Buildings and Bridges for Wind: A Practical Guide for ASCE 7 Users and Designers of Special Structures*, John Wiley and Sons, New York.
23. Ueda, T., Tanaka, T., and Matsushita, Y. (1998) Aerodynamic stabilization for super long-span suspension bridges. IABSE Symposium on Long Span and High Rise Structures, Japan.
24. Brown, W.C. (1996) Development of the deck for the 3300 m span Messina Crossing. 15th IABSE Congr. Rep., IABSE, Zurich, pp. 1019–1030.
25. Diana, G. (1993) Analytical and wind tunnel simulations for the aeroelastic design of the Messina Straits Bridge. Proceedings of International Seminar on Utilization of Large Boundary Wind Tunnel, Tsukuba, Japan, pp. 183–202.
26. Raggett, J.D. (1987) Stabilizing pair of winglets for slender bridge decks. Bridges and Transmission Line Structures, Sixth Structural Congress of ASCE, New York, pp. 292–302.
27. Arco, D.C. and Aparicio, Á.C. (1999) Improving suspension bridge wind stability with aerodynamic appendages. *Journal of Structural Engineering*, **125** (12), 1367–1375.
28. Kobayashi, H. and Nagaoka, H. (1992) Active control of flutter of a suspension bridge. *Journal of Wind Engineering and Industrial Aerodynamics*, **41** (3), 143–151.
29. Wilde, K. and Fujino, Y. (1998) Aerodynamic control of bridge deck flutter by active surfaces. *Journal of Engineering Mechanics*, ASCE, **124** (7), 718–727.
30. Kobayashi, H., Ogawa, R., and Taniguchi, S. (1998) Active flutter control of a bridge deck by ailerons. Proceedings of 2nd World Conference on Structural Control, Kobori ed., Kyoto, Japan, Vol. 3, pp. 1841–1848.
31. Wilde, K., Omenzetter, P., and Fujino, Y. (2001) Suspension of bridge flutter by active deck-flaps control system. *Journal of Engineering Mechanics*, **127** (1), 80–89.
32. Omenzetter, P., Wilde, K., and Fujino, Y. (2000) Suppression of wind-induced instabilities of a long span bridge by a passive deck-flaps control system Part I: formulation. *Journal of Wind Engineering and Industrial Aerodynamics*, **87** (1), 61–79.
33. Omenzetter, P., Wilde, K., and Fujino, Y. (2000) Suppression of wind-induced instabilities of a long span bridge by a passive deck-flaps control system Part II: Numerical simulations. *Journal of Wind Engineering and Industrial Aerodynamics*, **87** (1), 81–91.
34. Omenzetter, P., Wilde, K., and Fujino, Y. (2002) Study of passive deck flaps flutter control system on full bridge model. I: theory. *Journal of Engineering Mechanics*, **128** (3), 264–279.
35. Omenzetter, P., Wilde, K., and Fujino, Y. (2002) Study of passive deck flaps flutter control system on full bridge model. II: results. *Journal of Engineering Mechanics*, **128** (3), 280–286.
36. Kwon, S.D., Jung, M.S.S., and Chang, S.P. (2000) A new passive aerodynamic control method for bridge flutter. *Journal of Wind Engineering and Industrial Aerodynamics*, **86** (2–3), 187–202.
37. Larsen, A., Esdahl, S., Andersen, J.E., and Vejrum, T. (2000) Storebaelt suspension bridge-vortex shedding excitation and mitigation by guide vanes. *Journal of Wind Engineering and Industrial Aerodynamics*, **88** (2–3), 283–296.
38. Larsen, A., Savage, M., Lafrenière, A. *et al.* (2008) Investigation of vortex response of a twin box bridge section at high and low Reynolds numbers. *Journal of Wind Engineering and Industrial Aerodynamics*, **96** (6–7), 934–944.
39. Ge, Y.J., Yang, Y.X., and Cao, F.C. (2011) VIV sectional model testing and field measurement of Xihoumen suspension bridge with twin box girder. 13th International Conference on Wind Engineering, July 10–15, Netherlands.
40. Li, H., Laima, S., Ou, J. *et al.* (2011) Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements. *Journal of Wind Engineering and Industrial Aerodynamics*, **33** (6), 1894–1907.
41. Federal Highway Administration (2007) Wind Induced Vibration of Stay Cables, FHWA-HRT-05-084.
42. Hikami, Y. and Shiraishi, N. (1988) Rain-wind induced vibrations of cables in cable stayed bridges. *Journal of Wind Engineering and Industrial Aerodynamics*, **29** (1–3), 409–418.
43. Xu, Y.L., Li, Y.L., and Shum, K.M. (2004) Stonecutters Bridge: Investigation of Aerodynamic Forces on Stay Cables - Wind Tunnel Investigation, Report to Highways Department of Hong Kong, October.
44. Li, M.S., Liao, H.L., Chen, X., and He, X.D. (2003) Stonecutters Bridge: Rain-Wind-Induced Vibration Tests of Stay Cables, Report to Ove Arup & Partners Hong Kong Ltd, March.
45. Fujino, Y. and Yoshida, Y. (2002) Wind-induced vibration and control of Trans-Tokyo Bay Crossing Bridge. *Journal of Structural Engineering*, **128** (8), 1012–1025.
46. Hirai, S., Honda, A., Kato, H. *et al.* (1993) Aerodynamic stability of Trans-Tokyo Bay Highway Bridge. *Journal of Wind Engineering and Industrial Aerodynamics*, **49** (1–3), 487–496.

47. Nobuto, J., Fujino, Y., and Ito, M. (1988) A study on the effectiveness of TMD to suppress a coupled flutter of bridge deck. *Proceedings of Japan Society of Civil Engineering, Hiroshima, Japan*, **5** (5), 413–416.
48. Gu, M., Chang, C.C., Wu, W., and Xiang, H.F. (1998) Increase of critical flutter wind speed of long-span bridges using tuned mass dampers. *Journal of Wind Engineering and Industrial Aerodynamics*, **73** (2), 111–123.
49. Pourzeynali, S. and Datta, T.K. (2002) Control of flutter of suspension bridge deck using TMD. *Journal of Wind and Structures*, **5** (5), 407–422.
50. Kwon, S.D. and Park, K.S. (2004) Suppression of bridge flutter using tuned mass dampers based on robust performance design. *Journal of Wind Engineering and Industrial Aerodynamics*, **92** (11), 919–934.
51. Chen, X.Z. and Kareem, A. (2003) Efficacy of tuned mass dampers for bridge flutter control. *Journal of Structural Engineering, ASCE*, **129** (10), 1291–1300.
52. Xue, S.D., Ko, J.M., and Xu, Y.L. (2000) Control of torsional flutter and buffeting of long span bridges using liquid column damper, in *Proceedings of International Conference on Advances in Structural Dynamics* (eds Ko, J.M. and Xu, Y.L.), Elsevier, Oxford, pp. 1413–1420.
53. Korlin, R. and Starossek, U. (2007) Wind tunnel test of an active mass damper for bridge decks. *Journal of Wind Engineering and Industrial Aerodynamics*, **95** (4), 267–277.
54. Achkire, Y., Bossens, F., and Preumont, A. (1998) Active damping and flutter control of cable-stayed bridges. *Journal of Wind Engineering and Industrial Aerodynamics*, **74–76**, 913–921.
55. Pourzeynali, S. and Datta, T.K. (2005) Semiactive fuzzy logic control of suspension bridge flutter. *Journal of Structural Engineering*, **131** (6), 900–912.
56. Gu, M., Chen, S.R., and Chang, C.C. (2002) Parametric study on multiple tuned mass dampers for buffeting control of Yangpu Bridge. *Journal of Wind Engineering and Industrial Aerodynamics*, **89** (11–12), 987–1000.
57. Lin, Y.Y., Cheng, C.M., and Lee, C.H. (2000) A tuned mass damper for suppressing the coupled flexural and torsional buffeting response of long-span bridges. *Engineering Structures*, **22** (9), 1195–1204.
58. Shum, K.M., Xu, Y.L. and Guo, W.H. (2008) Wind-induced vibration control of long span cable-stayed bridges using multiple pressurized tuned liquid column dampers, *Journal of Wind Engineering and Industrial Aerodynamics*, **96** (2), 166–192.
59. Shum, K.M., Xu, Y.L., and Guo, W.H. (2006) Buffeting response control of a long span cable-stayed bridge during construction using semi-active tuned liquid column dampers. *Wind & Structures-An International Journal*, **9** (4), 271–296.
60. Zhou, H.J. and Xu, Y.L. (2007) Wind-rain-induced vibration and control of stay cables in a cable-stayed bridge. *Journal of Structural Control and Health Monitoring*, **14** (7), 1013–1033.
61. Irvine, H.M. (1981) *Cable Structures*, MIT Press, Cambridge, Massachusetts.
62. Xu, Y.L. and Wang, L.Y. (2003) Analytical study of wind-rain-induced cable vibration: SDOF model. *Journal of Wind Engineering and Industrial Aerodynamics*, **91** (1–2), 27–40.
63. Main, J.A. (2002) Modeling the Vibrations of a Stay Cable with Attached Damper, PhD thesis, The Johns Hopkins University.
64. Pacheco, B.M., Fujino, Y., and Sulekh, A. (1993) Estimation curve for modal damping in stay cables with viscous damper. *Journal of Structural Engineering*, **119** (6), 1961–1979.
65. Main, J.A. and Jones, N.P. (2002) Free vibrations of cable with attached damper I: Linear viscous damper. *Journal of Engineering Mechanics*, **128** (10), 1062–1071.
66. Butcher, J.C. (1987) *The Numerical Analysis of Ordinary Differential Equations-Runge-Kutta and General Linear Methods*, John Wiley & Sons, Inc., NY, USA.
67. Yamaguchi, H. (1990) Analytical study on growth mechanism of rain vibration of cables. *Journal of Wind Engineering and Industrial Aerodynamics*, **33** (1–2), 73–80.
68. Krenk, S. (2000) Vibrations of a taut cable with an external damper. *Journal of Applied Mechanics*, **67** (4), 772–776.
69. Christenson, R.E., Spencer, B.F., and Johnson, E.A. (2006) Experimental verification of smart cable damping. *Journal of Engineering Mechanics ASCE*, **132** (3), 268–278.
70. Xu, Y.L. and Zhou, H.J. (2007) Damping cable vibration for a cable-stayed bridge using adjustable fluid dampers. *Journal of Sound and Vibration*, **306** (1–2), 349–360.
71. Singh, M.P., Verma, N.P., and Moreschi, L.M. (2003) Seismic analysis and design with Maxwell dampers. *Journal of Engineering Mechanics*, **129** (3), 273–282.
72. Li, Z.Q., Xu, Y.L., and Zhou, L.M. (2006) Adjustable fluid damper with SMA actuators. *Smart Materials and Structures*, **15** (5), 1483–1492.
73. Krenk, S. and Høgsberg, J.R. (2005) Damping of cables by a transverse force. *Journal of Engineering Mechanics*, **131** (4), 340–348.